

Asymptotics of assemblies of nested cycles

Exact counts and fixed order marked saddle expansions

Part II: sharp Gumbel errors and optimal continuous calibration
for the largest outer component

Part I: Report 167

Part II: research manuscript prepared for Vladimir Reshetnikov

Part I: 3 October 2026 Part II: 8 October 2026 merged: 9 October 2026

Abstract

Let $P(z) = \prod_{m \geq 1} (1 + z^m/m)$ and $F(z, u) = \exp\{u(P(z) - 1)\}$. The labeled count $a_n = n![z^n]F(z, 1)$ is OEIS A308338, and the coefficient of u^k counts objects with k nonempty nested-cycle components, giving A392471. With $c = e^{-\gamma}$, we prove

$$a_n \sim \frac{e^{c(1/2 - \log 2) - 1} c^{1/4 - c/2}}{2\sqrt{\pi}} n! n^{c/2 - 3/4} e^{2\sqrt{cn}}.$$

More generally, every fixed order has a complex-uniform expansion in $n^{-1/2}$ with polynomial coefficients in $\log n$; we give the first two corrections explicitly. A whole-circle estimate controls all secondary arcs, and a Gaussian Taylor lemma gives the stated sharp logarithmic remainder without crossing a natural boundary.

For the component count, holomorphic normalization justifies differentiation and gives explicit logarithmic and constant corrections to the mean and variance. In particular the mean divided by $\sqrt{n}$ tends to $e^{-\gamma/2}$. We recover the established Gaussian regime with variance equivalent $\sqrt{cn}/2$ and prove a rounding-safe asymptotic inverse for the integer count sequence.

The component asymptotics and generic expansive-assembly limit laws are prior results. The contribution developed here is the explicit sequence-specific marked expansion and its refinements. The companion contains bounded exact integer and symbolic checks. No worldwide priority claim, complete beyond-all-orders transseries, or certified numerical asymptotic error bound is asserted.

[write] **Status and trust boundary, added 6 October 2026, batch 108.** This is bundle Report 167 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a308338-nested-cycle-assemblies** (provenance, the sources as the write read them, relation to the repository, the collected non-claims and the reading conventions in Section 1.1). It is unrefereed and not formalized; its author line reads “Report 167” and its PDF author field is empty. *What rests on what.* Everything analytic rests on the component estimate (1.4), credited to Greene and Knuth through Flajolet, Fusy, Gourdon, Panario and Pouyanne and cited, not reproved. Given it, the text proves the fixed-order expansion (Theorem 2.1), the moment expansions (Theorem 7.1), the Gaussian limit (Corollary 7.2, a known regime re-proved with this normalization) and the two-ceiling inverse (Theorem 8.1); no constant or onset is made effective and every decimal is uncertified. *Two statements of the literature are put on record, with proofs.* (i) Riedel’s January 2026 conjecture $E[X] \sim N\sqrt{n}$ for the number of components is *proved* by (7.8), with $N = e^{-\gamma/2} = 0.7493\dots$; this excludes the alternative he raised, a power n^M with M “close to but not equal to $1/2$ ”, and his numerical $N \approx 0.7719351399$ is not the constant but agrees to nine decimals with the finite ratio $\mathbb{E}K_{4500}/\sqrt{4500}$ (Remark 7.3). The source quotes the estimate without saying that it settles the conjecture. (ii) The general- C

covariance display of Erlihson and Granovsky (preprint (4.54), published (4.18), both read) is negative at $p = 1$, $C = 3$, $u = 1$ and contradicts Corollary 7.2 in this report’s own model; the write gives the corrected display and locates the dropped factor C (Remark 10.1). The source had already found the negative value. The write also adds Remark 8.2 (the inversion against the transseries volume), notes on the OEIS entries as of 6 October 2026, and one question (Section 11). Nothing was submitted to the OEIS or sent to any author.

[write] Part II, added 9 October 2026 (second write). Since its second write the report has two Parts. Part I (Sections 1–11) is Report 167 as written and independently checked; the abstract and the status note above describe it alone, and nothing of it changed except the `\part` line, this note and dated notes at the end of Section 1.1 and of Section 11. Part II (Sections 12–23 and Appendices A–B, labels `ncy:gb:`) is the manuscript *Sharp Gumbel Errors and Optimal Continuous Calibration for Random Assemblies* (8 October 2026, “Research manuscript prepared for Vladimir Reshetnikov”). It studies a statistic Part I does not treat, the size M_n of the *largest outer component* (a nested cycle of maximal size, not the longest single cycle), for every assembly with weights $p_k = c + O(1/k)$ and in particular for the nested-cycle assemblies of A308338. With τ the exact size saddle and $L = \log(c/\tau)$ it proves the sharp ordinary Gumbel error $d_K(F_n, G) \sim \kappa \tau L^2/(4c)$, $\kappa = (2 + \sqrt{5})e^{-(3+\sqrt{5})/2}$ (Proposition 17.1), that no affine normalization removes this order (Theorem 17.2), an explicit increasing calibration with error $\tau/(2e) + O(\tau/L)$, where $\tau/(2e)$ is optimal among all continuous distribution functions (Theorems 18.1 and 19.1), a lattice law with error $O(\tau/L)$ and an exponential-integral refinement with error $O(\tau^{3/2}L^3)$ (Theorems 16.2 and 20.1); for nested cycles the raw error is $(e^{\gamma/2}\kappa/16)\log^2 n/\sqrt{n}$ (Corollary 21.1). Its only analytic input for nested cycles is Part I’s credited component estimate (1.4). It answers none of Part I’s questions. Section 12 records its provenance, its relation to Part I, the checks, its non-claims and its letters, which are not Part I’s (in Part II $L = \log(c/\tau)$, a_k are Poisson means and B_j tilted size moments; in Part I $L = \log n$, a_n is A308338 and B_m is A007838).

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Scope and reading guide. Sections 1–2 specify the object and results. Sections 3–5 prove the fixed-order coefficient theorem. Sections 6–8 give corrections, probability laws, and inversion. Section 9 separates exact verification from numerical diagnostics. Section 10 records source coverage and limits. The report is self-contained after the explicitly credited component coefficient estimate (1.4).

Part I

Asymptotics of assemblies of nested cycles: exact counts and fixed order marked saddle expansions

1 Exact objects and established input

A labeled cycle on an m -element set has $(m-1)!$ possible cyclic orders and exponential generating function z^m/m . Choosing at most one cycle of each length therefore gives

$$P(z) = \prod_{m \geq 1} \left(1 + \frac{z^m}{m}\right) = \sum_{m \geq 0} B_m \frac{z^m}{m!}, \quad p_m = \frac{B_m}{m!}. \quad (1.1)$$

The integers B_m form A007838. A nonempty collection of cycles of distinct lengths has a unique decreasing-length nesting order. It is one nested component in Riedel's interpretation; there is no extra ordering or geometric-embedding choice. A set of such nonempty labeled components has marked exponential generating function

$$F(z, u) = \exp\{u(P(z) - 1)\}, \quad f_n(u) = [z^n]F(z, u). \quad (1.2)$$

Thus

$$T(n, k) = \frac{n!}{k!} [z^n](P(z) - 1)^k, \quad a_n = n! f_n(1) = \sum_{k=0}^n T(n, k). \quad (1.3)$$

The a_n are A308338, with $a_0 = 1$, and the published triangle A392471 starts at $n = 1$, $1 \leq k \leq n$. We use the natural empty extension $T(0, 0) = 1$ and $T(n, 0) = 0$ for $n > 0$ only for recurrences. In particular $T(n, 1) = B_n$ for $n \geq 1$.

Ilya Gutkovskiy entered A308338 and its defining EGF on 20 May 2019 [2]. Riedel's January 2026 note supplies the nested-cycle interpretation, the marked triangle, and recurrences [4, 3]. The component EGF is much older. The prior estimate we use is

$$p_m = c + \frac{c}{m} + O\left(\frac{\log m}{m^2}\right), \quad c = e^{-\gamma}, \quad m \longrightarrow \infty, \quad (1.4)$$

where γ is Euler's constant. Flajolet, Fusy, Gourdon, Panario, and Pouyanne [5, Section 3, equation (22)] attribute this estimate to Greene and Knuth. Their Proposition 1 gives the full component coefficient expansion, including periodic terms from roots of unity. We use (1.4) as a credited theorem; their theorem for P is not itself a coefficient theorem for the outer exponential F .

For a component spectrum $(j_1, \dots, j_n)$ with $\sum m j_m = n$, the number of objects is

$$n! \prod_{m=1}^n \frac{p_m^{j_m}}{j_m!}.$$

After division by a_n , this is the Gibbs assembly law with weights $p_m \sim c$, so its expansive exponent is 1. The weights are p_m , not B_m or p_m/m . The square-root component law and normal fluctuations lie within established expansive-assembly theory. Erlihson and Granovsky [6, Corollary 4.5] explicitly attribute the total-component CLT to their 2004 work [7]. We derive the precise normalization and lower-order terms directly below.

1.1 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 108.** *Provenance.* This report is Report 167 of the session bundle that ProveIt’s docs/incoming intake processed as batch 108: the archive `Nested_Cycle_Assemblies_Asymptotics_and_Inverses_Source.zip` (938,156 bytes, SHA-256 `ef59faff7aba...eaebc0b63`, 21 files at the archive root and in its `companion/` and `data/` folders), arrival commit `60f54ea06`, placed in `602e5bd0f`. The text is the delivered `Report167.tex` (908 lines, dated 3 October 2026), shipped as `article.tex`. The delivered 15-page PDF and the checksum manifest `SHA256SUMS` (20 entries, verified at placement and again at the write) are not shipped, and the delivered README is replaced by the report README; all three are retrievable from the arrival commit. The package records no ProveIt commit (it cites nothing in the repository), no “prepared for private review” line and no personal data; the word “private” occurs only in the delivered README’s “private fresh format” (a $\text{\TeX}$ format file). It is a single-source report, so no merge choice was made. The write prefixed the 62 delivered labels with `ncy:` and updated the 60 references to them; it added this subsection, the notes marked [write], Remarks 7.3, 8.2 and 10.1, labels on the items of Section 11 and one further item there. No statement, proof or number of the source was changed: the added remarks are the last statements of their sections, the added displays are unnumbered, and the added item follows the last delivered one, so every theorem, section and equation of the source keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A308338 [2], live revision #21 (5 October 2026); the source froze revision #17 (15 January 2026). Revisions #18–#20, by Alois P. Heinz on 5 October 2026, add a Maple program and a b-file for $0 \leq n \leq 445$ (#21 is the OEIS server installing that b-file); the 22 data terms are unchanged. (Wording corrected after the independent check of 6 October 2026; the first wording read “Revisions #18–#21, by Alois P. Heinz”.) Riedel’s comment identifying the terms as nested cycle partitions and his formula $a(n) = \sum_k T(n, k)$ date from 14 January 2026. The entry states no asymptotic and no conjecture.
- OEIS A392471 [3], live revision #42 (5 October 2026); the source froze revision #37 (25 January 2026). On 5 October 2026 Heinz inserted the column $k = 0$ (the entry’s `%E` line reads “Column k=0 inserted by Alois P. Heinz, Oct 05 2026”), so the entry now has offset 0 and rows from $n = 0$, and he added a Maple program and a b-file of rows $0 \leq n \leq 150$. The delivered sentence of Section 1 that the published triangle “starts at $n = 1$, $1 \leq k \leq n$ ” describes revision #37; the entry now uses the same extension $T(0, 0) = 1$, $T(n, 0) = 0$ ($n > 0$) as this report. Its link to Riedel’s original note is titled “Nested Cycle Partitions: definitions and basic recurrences”; the PDF behind it is titled “Nested Cycle Partitions: a conjecture”. The entry itself contains no conjecture.
- OEIS A007838 [1], revision #74 (13 March 2026), unchanged; its formula section states (1.4) in the form $a(n) \sim n!(e^{-g} + e^{-g}/n + O((\log n)/n^2))$ and its reference section cites Greene and Knuth.
- The three b-files agree with `data/exact_data_100.json` for $0 \leq n \leq 100$ (A308338, A007838) and for all rows $0 \leq n \leq 100$ of the triangle (the write’s comparison).
- Riedel’s note [4] in both versions: the revised one (author-hosted, page footers dated 24.01.26) and the original linked from A392471 (footers dated 13.01.26), six pages each; the paragraph “The conjecture” on p. 3 is the same in both (Remark 7.3).
- Erlihson and Granovsky [6]: the preprint arXiv:math/0507343v3 (19 June 2007), Section 4 from (4.37) to (4.55) and Section 5 around (5.121), and the published text (numdam), pp. 925–927 and 936–938 (Remark 10.1).
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:prop:factorial-core`, `p0:def:three-inverses` and `p0:thm:staircase`, read for Remark 8.2.
- Not read by the write: the hybrid paper [5] (the source read its Section 3 and Proposition 1), the Greene–Knuth book and the 2004 paper of Erlihson and Granovsky [7] (the source read neither). `data/SOURCE_PROVENANCE.json` records what the source inspected.

What was checked. The batch-108 dossier read the manuscript in full and found no false claim. It checked (1.4) at $m = 100$ ($p_{100} = 0.56678641$ against $c + c/100 = 0.56707408$, a difference within $\log m/m^2$), the counts, the exact mean and the exact variance at $n = 100$ against (2.3) and Theorem 7.1, read Riedel’s conjecture paragraph, verified the literal substitution in the Erlihson–Granovsky preprint, and replayed the companion on copies (all four frozen outputs reproduced). At the write the companion was rerun from a fresh extraction of the archive and from the shipped files (note at the end of Section 9), and $f_n(1)$, $\mathbb{E}K_n$ and $\text{Var } K_n$ were evaluated in double precision for $n \leq 40000$ by recurrences with positive terms only (p_m from the cycle product of (9.1) divided by $m!$, $f_n(1)$ from the second recurrence of (9.3), and the moments from $[z^n]F(z, 1)(P(z) - 1)$ and $[z^n]F(z, 1)(P(z) - 1)^2$), agreeing with the exact rational values at $n = 100$ ($\mathbb{E}K_{100} = 8.219543$, $\text{Var } K_{100} = 4.016577$). The residuals are consistent with Theorem 2.1 at $J = 1$ and with Theorem 7.1 including M_1 and V_1 : at $n = 40000$, $\sqrt{n}(a_n/(Cn!n^{c/2-3/4}e^{2\sqrt{cn}}) - 1) = 9.682$ against $R_1(\log n; 1) = 9.344$, $\sqrt{n}$ times the residual of (7.2) is 12.92 against $M_1(\log n) = 12.69$, and $\sqrt{n}$ times the residual of (7.3) is 16.97 against $V_1(\log n) = 16.53$; at $n = 1000$ the same pairs are 5.43/4.71, 6.74/6.13 and 8.38/7.27, so the gaps shrink as the next order predicts. These are observations, not certificates.

Relation to the repository. Before batch 108 no file of the repository named A308338, A392471 or A007838, nested cycle partitions or Riedel’s note, and none does outside this report and the collection catalogue. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The neighbouring reports `a064856-stirling-catalan-transforms` (Part II: cycle-weighted permutation mixtures) and `a301746-divisor-weighted-asymptotics`, `a022629-distinct-partition-norms`, `a271619-strict-twice-partitions` (Gibbs partitions after Granovsky, Stark and Erlihson) treat other models; none uses the Erlihson–Granovsky display corrected in Remark 10.1, and none needs a reciprocal note. The inversion of Section 8 is compared with the transseries volume in Remark 8.2.

[write] What is not claimed, collected from the source. The expansion of Theorem 2.1 holds at every fixed order only: no convergence, optimal truncation, Stokes phenomenon or complete beyond-all-orders description is claimed, and constants need not be uniform when the order grows. The decimal value of C is an uncertified floating evaluation; no interval enclosure of C is supplied, and no implied constant or onset (in particular no certified M_J or threshold in Theorem 8.1) is made effective. The calculator (6.3) is formal: its contour is not identified with a Bessel contour and no convergence is claimed. The square-root law, the exact limit (7.8) and the normal fluctuations belong to the established expansive-assembly regime; Corollary 7.2 is “not a new universality assertion”, and no local limit theorem or distribution distance bound is inferred. The finite and symbolic checks prove no asymptotic statement; the B , B_3 checks verify the arithmetic pieces, not the infinite-tail summations. Source coverage is bounded: no worldwide priority is claimed, “No claim is made that no other work contains overlapping results”, the Greene–Knuth book and the 2004 paper were not inspected, and no OEIS entry was amended. Byte identity of rebuilds is toolchain-specific. The write adds: the table and the identification of $n = 4500$ in Remark 7.3 are floating evaluations (checked against exact and 256-bit fixed-point values) and an inference about how Riedel’s number arose, not a statement of his; the corrected display of Remark 10.1 is derived from the conditioning computation, not from a re-audit of the whole proof of Erlihson and Granovsky.

Reading conventions (letters the source reuses, and letters of the two outside sources discussed by the write, with the tempting false readings; no symbol of the source was renamed; ϱ and $M_\varrho(n)$ are the write's).

Symbol	Meaning here (where)	Not to be read as
B_m, p_m	component counts (A007838) and $B_m/m!$	the constants B, B_3 of (3.4); the calculator $\mathcal{B}_\beta$; Erlihson–Granovsky's b_r and exponent p
c, A, D, E	$e^{-\gamma}$; constants of (2.1), (3.6)	OEIS A-numbers; $\mathbb{E}$; the residual $E_{n,J}$
$C(u), C$	amplitude (2.1), (2.3)	$C_0, C_{j,k,m}$ (proof constants); Erlihson–Granovsky's C in $a_k \sim Ck^{p-1}$
$u, v = cu$	marking variable; cu	Erlihson–Granovsky's scaled size u ; v of <code>p0:prop:factorial-core</code>
L	$\log n$ in $R_j(L; u), M_1(L), V_1(L)$; $\log y$ in Section 8	$L_n(u)$ (leading term); the target L of <code>p0:prop:factorial-core</code>
$Q(X), Q_j, X_n$	polynomials of (3.6), (5.2); contour bound	Riedel's $Q(z, u) = F(z, u)$ and $Q_k(z)$; Riedel's $X = K_n$
K_n, K, K_0	component count; constant in the proof of Theorem 8.1; $\log(C\sqrt{2\pi})$	
$H_{j,k}, H_j$	Taylor factor (5.5); harmonic numbers	
ℓ, ℓ_c, ℓ_r	$\log s$; $(\log c - L)/2$; coefficients of $\log P$ (Section 9)	
β	$c/2 - 3/4$ (Section 8)	the free exponent in $\mathcal{B}_\beta$
$\kappa, \kappa', h, d, \delta$	loss constants, window $a^{3/2} \log n$ (Section 4); bound and $n^{-1/4}$ in Lemma 5.1	the volume's $\kappa_{\text{vol}}, d_{\text{vol}}, h$; Riedel's κ_n ; Erlihson–Granovsky's $\delta = 1/r_N$
p, q	$2J + 2, (J + 1)/2$ (Section 8); $q = Q(\ell_c)$ in (7.6)	p_m ; Erlihson–Granovsky's exponent p and number q of strata
$N(y), M_J, M_1$	first passage; constant of Theorem 8.1; mean correction	Riedel's constant N and exponent M ; Erlihson–Granovsky's size N ; the write's $M_\varrho(n)$
$T(n, k), t$	A392471 (with $T(n, 0)$); $z = e^{-t}$	

[write] **Part II and the scope of this Part, 9 October 2026.** Part I treats the counts a_n , the triangle $T(n, k)$, the number K_n of components and the inverse $N(y)$; it says nothing about the sizes of the components, and in particular nothing about the largest one. Part II (Section 12) treats that statistic, the size M_n of the largest outer component, with the same model and the same credited estimate (1.4), which there takes the form of its refined weight hypothesis (14.10) with $c = a_* = e^{-\gamma}$. It read Part I at the blob 07022425b718, which is Part I's `article.tex` before this write, and uses Part I for the model, its counts and its credits only, not as a premise of any proof. The relation to the repository stated above is unchanged by Part II: no other report treats largest components of these assemblies, and none needs a reciprocal note.

2 The marked coefficient theorem

Fix a sufficiently small open complex disc about $u = 1$. Choose $\log(cu)$ and $\sqrt{cu}$ to be real on positive real u , and set

$$A = \frac{1}{2} - \log 2, \quad C(u) = \frac{e^{u(cA-1)}(cu)^{1/4-cu/2}}{2\sqrt{\pi}}. \quad (2.1)$$

All statements about fixed order mean that the order is fixed as $n \rightarrow \infty$; constants need not be uniform when that order grows.

Theorem 2.1 (Uniform fixed-order expansion). *For every fixed integer $J \geq 0$ there are polynomials $R_j(L; u)$ of degree at most $2j$ in L , with coefficients holomorphic near 1, such that*

$$f_n(u) = C(u)n^{cu/2-3/4}e^{2\sqrt{cun}} \left\{ 1 + \sum_{j=1}^J n^{-j/2} R_j(\log n; u) + E_{n,J}(u) \right\}, \quad (2.2)$$

$$E_{n,J}(u) = O\left(n^{-(J+1)/2}(1 + \log n)^{2J+2}\right).$$

The estimate is uniform on every sufficiently small fixed closed disc about 1. The exact normalized residual $E_{n,J}$ is holomorphic on an open disc containing any smaller disc on which derivatives are taken, and each fixed derivative obeys the same order bound there.

Taking $u = 1$, $J = 0$ gives the equivalent

$$a_n \sim Cn!n^{c/2-3/4}e^{2\sqrt{cn}}, \quad C = \frac{e^{c(1/2-\log 2)-1}c^{1/4-c/2}}{2\sqrt{\pi}}. \quad (2.3)$$

For orientation only, $C \approx 0.0947779253874791584250013230745944160$; this is an uncertified floating evaluation of the exact expression. The logarithmic term of P supplies the power $n^{c/2}$ in (2.3). Discarding that term would give an incorrect equivalent.

The theorem is an all-fixed-orders algebraic expansion with logarithmic polynomials. It makes no claim about convergence, optimal truncation, Stokes phenomena, or a complete description of terms beyond all orders.

3 Local component expansion in a right half plane

We first prove the needed local expansion without continuing P through its natural boundary. For $\Re t > 0$, isolating the factor $m = 1$ gives the absolutely convergent identity

$$\begin{aligned} \log P(e^{-t}) &= \log(1 + e^{-t}) + \text{Li}_1(e^{-t}) - e^{-t} \\ &+ \sum_{k \geq 2} \frac{(-1)^{k+1}}{k} \{\text{Li}_k(e^{-kt}) - e^{-kt}\}. \end{aligned} \quad (3.1)$$

Fix a desired order H and choose an integer $K > H + 2$. The tail $k > K$ can be written as a double sum over $k > K$, $m \geq 2$. For $0 \leq h \leq H + 1$, its h th derivative in t is bounded absolutely and uniformly on the closed right half plane by

$$\sum_{k > K} k^{h-1} \sum_{m \geq 2} m^{h-k} < \infty. \quad (3.2)$$

Indeed $k - h > 1$ and the inner sum decreases exponentially as k grows. Termwise differentiation up to that order is consequently valid. The integral remainder along the segment from 0 to t gives a Taylor polynomial for the tail with error $O(|t|^{H+1})$, uniformly in the closed right half plane near 0.

Each retained polylogarithm has the classical finite-order expansion

$$\text{Li}_k(e^{-x}) = \frac{(-x)^{k-1}}{(k-1)!} (H_{k-1} - \log x) + \sum_{\substack{r \geq 0 \\ r \neq k-1}} \zeta(k-r) \frac{(-x)^r}{r!}, \quad (3.3)$$

where $H_j = \sum_{m=1}^j 1/m$ and $H_0 = 0$. For a fixed finite set of k , truncations and their remainders are uniform in every closed subsector of $\Re t > 0$. Combining them with (3.2) gives each fixed order of $\log P(e^{-t})$. At every positive power t^r , the coefficient is affine in $\log t$. Its logarithmic coefficient comes only from $k = r + 1$ in (3.1) and is

$$-\frac{(r+1)^{r-1}}{r!} \quad (r \geq 1).$$

The constant after removing $-\log t$ is $-\gamma$, by the convergent Euler product with the harmonic term subtracted. Explicitly,

$$\begin{aligned} \log P(e^{-t}) = & -\log t - \gamma + t(-\log t + A) + t^2 \left(-\frac{3}{2} \log t + B \right) \\ & + t^3 \left(-\frac{8}{3} \log t + B_3 \right) + O(|t|^4(1 + |\log t|)), \end{aligned} \quad (3.4)$$

$$B = \frac{41}{24} - \frac{\zeta(2)}{2} - \frac{3}{2} \log 3, \quad B_3 = \frac{49}{12} - \frac{\pi^2}{4} + \frac{\zeta(3)}{3} - \frac{8}{3} \log 4.$$

For completeness, the t^2 constant contributions from the isolated prefactor, $k = 2$, $k = 3$, and $k \geq 4$ are, respectively,

$$-\frac{5}{12}, \quad \frac{3}{2}, \quad \frac{3}{4} - \frac{3}{2} \log 3, \quad -\frac{\zeta(2)}{2} - \frac{1}{8}.$$

The last follows by summing a rational series:

$$\begin{aligned} \frac{1}{2} \sum_{k \geq 4} (-1)^{k+1} k(\zeta(k-2) - 1) &= -\frac{1}{2} \sum_{m \geq 2} \frac{4m+3}{m(m+1)^2} \\ &= -\frac{\zeta(2)}{2} - \frac{1}{8}. \end{aligned}$$

For B_3 , the corresponding contributions (now separating $k = 4$) are $1/6$, $-13/18$, $9/4$, $20/9 - (8/3) \log 4$, and

$$-\frac{1}{6} \sum_{m \geq 2} \frac{25m^2 + 39m + 16}{m(m+1)^3} = \frac{1}{6} - \frac{3}{2} \zeta(2) + \frac{1}{3} \zeta(3).$$

These calculations supply the constants rather than attributing them to a short displayed formula in the component source.

Exponentiation to any prescribed fixed order yields

$$P(e^{-t}) = \frac{c}{t} + c(-\log t + A) + \sum_{j=1}^M t^j U_j(\log t) + O(|t|^{M+1}(1 + |\log t|)^{M+2}), \quad (3.5)$$

where $\deg U_j \leq j + 1$. To obtain the stated error, first expand $\log P$ through degree $M + 1$, with error $O(|t|^{M+2}(1 + |\log t|))$, and then exponentiate. The largest omitted logarithmic degree comes from $M + 2$ copies of $t \log t$. Define

$$\begin{aligned} D &= \log 2 - 2, \quad E = B + A^2/2, \quad Q(X) = X^2/2 + DX + E, \\ Z(X) &= \frac{(-X + A)^3}{6} + (-X + A) \left(-\frac{3}{2}X + B \right) - \frac{8}{3}X + B_3. \end{aligned} \quad (3.6)$$

Then $U_1 = cQ$ and $U_2 = cZ$. All these are sector expansions inside the right half plane. No analytic continuation of P across $|z| = 1$ has been used.

4 The complete circle and complex saddle

The global step comes from the prior coefficient estimate rather than from a claim that the positive real singularity is the only singularity. From (1.4), absolute summation of $|p_m - c|e^{-am}$, including the bounded $m = 0$ adjustment, gives uniformly for $0 < a < 1$, $|y| \leq \pi$,

$$P(e^{-a-iy}) = \frac{c}{1 - e^{-a-iy}} + O(1 + \log(1/a)) = \frac{c}{a + iy} + O(1 + \log(1/a)). \quad (4.1)$$

For the second equality, $(1 - e^{-t})^{-1} - t^{-1}$ is bounded on $0 \leq \Re t \leq 1$, $|\Im t| \leq \pi$ after filling its removable value at 0. Thus (4.1) includes every secondary root-of-unity arc.

Put $v = cu$ and $s = \sqrt{v/n} = a + ib$, where $a > 0$. Coefficient extraction on $|z| = e^{-a}$, with $z = e^{-t}$, gives

$$f_n(u) = \frac{1}{2\pi i} \int_{a-i\pi}^{a+i\pi} \exp\{u(P(e^{-t}) - 1) + nt\} dt. \quad (4.2)$$

The sign from $dz/z = -dt$ is canceled by reversal of the circle parameter. The principal phase is $\Phi(t) = v/t + nt$, with saddle s . Since $v = ns^2$, direct algebra gives the exact loss identity

$$\Re \Phi(a + iy) = 2na - na \frac{(y - b)^2}{a^2 + y^2}. \quad (4.3)$$

Uniformly on the small marking disc, $a \asymp n^{-1/2}$ and $|b|/a$ is bounded. Take $h = a^{3/2} \log n$. Outside $|y - b| < h$, the loss in (4.3) is at least $\kappa(\log n)^2$ for a positive constant κ . To see this, when $|y| \leq C_0 a$ the denominator is $O(a^2)$ and $nah^2/a^2 \asymp (\log n)^2$; when $|y| > C_0 a$ with C_0 sufficiently large, the fraction is bounded below, and the loss is of order $\sqrt{n}$.

The error from (4.1) costs only $O(\log n)$ in the real exponent, with u bounded. The entire off-saddle contribution is therefore

$$O\left(e^{2na - \kappa'(\log n)^2}\right) \quad (4.4)$$

for a uniform $\kappa' > 0$. The modulus of the leading term in (2.2) is e^{2na} times a quantity bounded above and below by fixed powers of n . Hence (4.4), relatively to that term, is smaller than every fixed inverse power of n . This proves the needed suppression for the whole circle, including the natural-boundary points, rather than relying only on positivity or aperiodicity.

5 The local Gaussian expansion and its remainder

On $|t - s| \leq h$, t remains in a fixed right-half-plane subsector and $|t| \asymp n^{-1/2}$. Equation (3.5) gives, for any fixed J ,

$$\begin{aligned} e^{u(P(e^{-t})-1)} &= e^{u(cA-1)} t^{-v} e^{v/t} \\ &\times \left\{ \sum_{j=0}^J t^j Q_j(\log t; u) + O(|t|^{J+1}(1 + |\log t|)^{2J+2}) \right\}. \end{aligned} \quad (5.1)$$

Here $Q_0 = 1$, and all Q_j are defined by the formal identity

$$\exp \left(u \sum_{j \geq 1} t^j U_j(X) \right) = \sum_{j \geq 0} t^j Q_j(X; u). \quad (5.2)$$

A product of r positive-degree factors with total t -degree j has logarithmic degree at most $j+r \leq 2j$; therefore $\deg_X Q_j \leq 2j$. Truncating the ordinary exponential gives the error in (5.1), since $t(1+|\log t|)^2 \rightarrow 0$ on the segment. The component error in (3.5) is no larger. In particular,

$$Q_1 = vQ, \quad Q_2 = vZ + \frac{v^2}{2}Q^2. \quad (5.3)$$

The next lemma records the integrated Taylor remainder carefully; replacing powers of the integration variable by endpoint bounds would lose unnecessary powers of $\log n$.

Lemma 5.1 (Uniform Gaussian Taylor remainder). *Set $\delta = n^{-1/4}$, $\lambda = \sqrt{vn}$, and use the actual vertical contour parametrization*

$$t = s + in^{-3/4}x = s(1 + i\delta x/\sqrt{v}), \quad |x| \leq X_n = (\Re\sqrt{v})^{3/2} \log n.$$

For fixed nonnegative integers j, k, S , the integral of the monomial $t^{j-v}(\log t)^k$ times $e^{nt+v/t}$ has an expansion, after removing its common leading scale, in powers $n^{-r/2}$, $0 \leq r \leq S$. Its relative-to-that-scale remainder is $O(n^{-(S+1)/2}(1+|\log s|)^k)$, uniformly in u on a sufficiently small fixed closed disc. Each coefficient is polynomial in $\log s$ of degree at most k and holomorphic in u .

Proof. The exact phase, with no truncated exponential yet, is

$$nt + \frac{v}{t} = 2\lambda - \frac{x^2}{\sqrt{v}} + \frac{i\delta x^3}{v(1 + i\delta x/\sqrt{v})}. \quad (5.4)$$

Write $\ell = \log s = (\log v - \log n)/2$ and remove s^{j-v} from the monomial. The residual analytic factor multiplying the Gaussian is

$$\begin{aligned} H_{j,k}(\delta, x; v, \ell) &= (1 + i\delta x/\sqrt{v})^{j-v} [\ell + \log(1 + i\delta x/\sqrt{v})]^k \\ &\quad \times \exp\left(\frac{i\delta x^3}{v(1 + i\delta x/\sqrt{v})}\right). \end{aligned} \quad (5.5)$$

In a Taylor expansion in δ , hold ℓ fixed; substitute its stated value afterwards. Let $d > 0$ be a uniform lower bound for $\Re(v^{-1/2})$ on the marking disc. For every fixed derivative order m and all sufficiently large n , one has, for $0 \leq \xi \leq \delta$ and $|x| \leq X_n$,

$$|\partial_\delta^m H_{j,k}(\xi, x; v, \ell)| \leq C_{j,k,m}(1 + |\ell|)^k P_{j,k,m}(|x|)e^{dx^2/2}, \quad (5.6)$$

with a fixed polynomial $P_{j,k,m}$ having nonnegative coefficients. Indeed, $|\xi x/\sqrt{v}| \leq 1/2$ eventually. Differentiation of the rational exponent and logarithm gives fixed polynomial bounds in $|x|$, with denominators uniformly separated from zero. The exponential modulus is at most $\exp(C\delta|x|^3)$, and $C\delta X_n \leq d/2$ eventually. The complex power and its fixed number of derivatives are bounded in the same way by polynomials. Finite product and chain rules prove (5.6).

Expand (5.5) through degree $2S+1$. The integral Taylor remainder, multiplied by $e^{-x^2/\sqrt{v}}$ and integrated, is at most

$$C\delta^{2S+2}(1 + |\ell|)^k \int_{\mathbb{R}} P_{j,k,2S+2}(|x|)e^{-dx^2/2} dx = O(n^{-(S+1)/2}(1 + |\ell|)^k). \quad (5.7)$$

This uses Gaussian moments, not the bound $|x| \leq X_n$ for the polynomial factor, so it introduces no extra powers of $\log n$. The identity $H_{j,k}(-\delta, -x) = H_{j,k}(\delta, x)$ implies that the coefficient of δ^m

has parity m in x . All odd coefficients integrate to zero on the symmetric centered segment, also for complex u because its center is the actual saddle s . The remaining even coefficients can be integrated over $\mathbb{R}$: the tails beyond $X_n \asymp \log n$ are smaller than every fixed inverse power of n .

The moments used are

$$\int_{\mathbb{R}} x^{2r} e^{-x^2/\sqrt{v}} dx = \sqrt{\pi} \frac{(2r-1)!!}{2^r} v^{(2r+1)/4}, \quad \int_{\mathbb{R}} x^{2r+1} e^{-x^2/\sqrt{v}} dx = 0, \quad (5.8)$$

with $(-1)!! = 1$. They hold first for real positive v by substitution and then throughout the disc by holomorphy of dominated Gaussian integrals. This use of continuation concerns the Gaussian integral only. Polynomial dependence on ℓ and holomorphic dependence on u are preserved by the finite Taylor and moment operations. $\square$

Proof of Theorem 2.1. For a monomial in Q_j , apply Lemma 5.1 with $S = J - j$. Its s^j contributes $n^{-j/2}$, and its logarithmic degree is at most $2j$. Its total relative error is therefore

$$O(n^{-(J+1)/2} (1 + \log n)^{2j}).$$

The remainder in (5.1), integrated absolutely with the local Gaussian, contributes $O(n^{-(J+1)/2} (1 + \log n)^{2J+2})$ relative to the leading scale. Equation (4.3), or (5.4) with its small correction, supplies a fixed Gaussian majorant; t^{-v}/s^{-v} is bounded on the segment. Equation (4.4) disposes of the rest of the circle.

The Jacobian and leading Gaussian integral give precisely

$$\frac{n^{-3/4}}{2\pi} e^{u(cA-1)} s^{-v} e^{2\lambda} \sqrt{\pi} v^{1/4} = C(u) n^{v/2-3/4} e^{2\sqrt{vn}}.$$

Every even correction coming from Q_j has logarithmic degree at most $2j$. At total order $n^{-r/2}$, $j \leq r$, proving the degree bound $2r$ and the coefficient holomorphy.

Finally, let $L_n(u) = C(u) n^{cu/2-3/4} e^{2\sqrt{cun}}$. It is holomorphic and nonzero on a small fixed open disc. Since $f_n(u)$ is a polynomial, the exact expression

$$E_{n,J}(u) = \frac{f_n(u)}{L_n(u)} - 1 - \sum_{j=1}^J n^{-j/2} R_j(\log n; u) \quad (5.9)$$

is holomorphic there. The proof gives its bound on every sufficiently small closed disc. Choose two concentric discs whose closures are inside the open disc; Cauchy's derivative estimates transfer the bound on the larger disc to all fixed derivatives on the smaller disc, without changing its powers of n or $\log n$. The contour radius $a = \Re \sqrt{cu/n}$ need not vary holomorphically: it is a device for proving a bound on the already holomorphic expression (5.9). $\square$

6 Explicit corrections and a fixed order calculator

Retain $v = cu$, $\ell = (\log v - L)/2$ and Q, Z, D from (3.6). The first two correction polynomials are

$$R_1(L; u) = v^{3/2} Q(\ell) - \frac{4(v-1)^2 - 1}{16\sqrt{v}} \quad (6.1)$$

and

$$\begin{aligned}
R_2(L; u) = & \frac{[4(v-1)^2 - 1][4(v-1)^2 - 9]}{512v} \\
& + v \left\{ \frac{1 - 4(v-2)^2}{16} Q(\ell) + \frac{v-2}{2}(\ell + D) - \frac{1}{4} \right\} \\
& + v^2 Z(\ell) + \frac{v^3}{2} Q(\ell)^2.
\end{aligned} \tag{6.2}$$

In particular the coefficient of L^2 in R_1 is $v^{3/2}/8$.

Here is a direct Gaussian check of the first correction. The unperturbed amplitude t^{-v} has order- δ^2 integrand

$$-\frac{v+1}{2}x^2 + \frac{v+1}{v^{3/2}}x^4 - \frac{1}{2v^2}x^6.$$

The normalized Gaussian moments are $\sqrt{v}/2$, $3v/4$, and $15v^{3/2}/8$, giving $-[4(v-1)^2 - 1]/(16\sqrt{v})$. The tQ_1 term contributes $v^{3/2}Q(\ell)$, proving (6.1).

For clarity, the four parts of (6.2) have separate origins. The unperturbed fourth-order term gives its first line. The order- δ^2 correction to tQ_1 gives its second line. The leading Gaussian integral of t^2Q_2 gives its last line, by (5.3). Each follows from (5.5) and the exact moments (5.8); the companion repeats these identities symbolically. This makes R_2 a finite algebraic consequence of the same local calculation, rather than an additional asymptotic assumption.

For general fixed order, equations (3.1), (5.2), and (5.5) are an algorithm using only finitely many coefficients and Gaussian moments. An equivalent compact formal calculator for a monomial $t^{-\beta}$ is

$$\begin{aligned}
\mathcal{B}_\beta(v, n) = & \left(\frac{n}{v}\right)^{(\beta-1)/2} \frac{e^{2\sqrt{vn}}}{\sqrt{4\pi\sqrt{vn}}} \\
& \times \sum_{r \geq 0} \frac{(-1)^r \prod_{l=1}^r [4(\beta-1)^2 - (2l-1)^2]}{r!(16\sqrt{vn})^r}.
\end{aligned} \tag{6.3}$$

For $t^{j-v}(\log t)^k$, replace the monomial by $(-\partial_\beta)^k \mathcal{B}_\beta(v, n)$ at $\beta = v - j$, holding v fixed, multiply by $e^{u(cA-1)}$, and retain terms with $j + r \leq J$. Equation (6.3) is the dominant-saddle formal series familiar from the modified Bessel function. Its use here is a coefficient calculator: the original full contour has not been identified with a Bessel contour, and no convergence of the formal series is claimed. The remainder proof is the Gaussian argument of Section 5.

7 Refined component moments and the Gaussian limit

Let K_n be the number of nested components in an object chosen uniformly from the a_n labeled objects. Its probability generating function is

$$\mathbb{E}u^{K_n} = \frac{f_n(u)}{f_n(1)}.$$

Since $f_n/L_n \rightarrow 1$ uniformly near $u = 1$, the ratio is nonzero on a fixed smaller disc for all sufficiently large n and has the analytic logarithm tending to zero there. The theorem and finite logarithmic expansion imply

$$\log f_n(u) = \log L_n(u) + n^{-1/2}R_1(\log n; u) + O(n^{-1}(1 + \log n)^4), \tag{7.1}$$

with the same order for any fixed number of derivatives on a smaller disc. This follows by applying Cauchy's estimates to the exact normalized residual before differentiation. Thus no real-axis asymptotic is being differentiated without justification.

Theorem 7.1 (Mean and variance). *As $n \rightarrow \infty$,*

$$\mathbb{E}K_n = \sqrt{cn} + \frac{c}{2} \log n + m_0 + O(n^{-1/2}(1 + \log n)^2), \quad (7.2)$$

$$m_0 = -\frac{3}{4} - c \log 2 + \frac{c\gamma}{2},$$

$$\text{Var } K_n = \frac{\sqrt{cn}}{2} + \frac{c}{2} \log n + v_0 + O(n^{-1/2}(1 + \log n)^2), \quad (7.3)$$

$$v_0 = -1 - c \left(\log 2 + \frac{1}{2} - \frac{\gamma}{2} \right).$$

The next corrections are respectively $n^{-1/2}M_1(\log n)$ and $n^{-1/2}V_1(\log n)$, after which both errors are $O(n^{-1}(1 + \log n)^4)$, where

$$M_1(L) = c\partial_v \mathcal{R}_1(L, v)|_{v=c}, \quad V_1(L) = (c\partial_v + c^2\partial_v^2)\mathcal{R}_1(L, v)|_{v=c}. \quad (7.4)$$

Here $\mathcal{R}_1(L, v)$ is the right side of (6.1) with $\ell = (\log v - L)/2$, and L is held fixed.

Proof. The cumulant operator for marking by u is $\mathcal{D} = u\partial_u = v\partial_v$. Consequently the mean and variance are $\mathcal{D} \log f_n|_{u=1}$ and $\mathcal{D}^2 \log f_n|_{u=1}$. The logarithm of the leading term is

$$\log L_n(u) = 2\sqrt{vn} + \left(\frac{v}{2} - \frac{3}{4}\right) \log n + u(cA - 1) + \left(\frac{1}{4} - \frac{v}{2}\right) \log v - \log(2\sqrt{\pi}). \quad (7.5)$$

Differentiating this explicit holomorphic function and using $\log c = -\gamma$ gives (7.2)–(7.3). Differentiating the next term in (7.1) gives (7.4); its residual is controlled as just explained. $\square$

An entirely explicit form of these first refined polynomials is also useful. Write $q = Q(\ell_c)$, $q' = \ell_c + D$, and $\ell_c = (\log c - L)/2$. Then

$$M_1(L) = c^{3/2} \left(\frac{3}{2}q + \frac{1}{2}q' \right) - \frac{3}{8}c^{3/2} + \frac{1}{4}c^{1/2} + \frac{3}{32}c^{-1/2}, \quad (7.6)$$

$$V_1(L) = c^{3/2} \left(\frac{9}{4}q + \frac{3}{2}q' + \frac{1}{4} \right) - \frac{9}{16}c^{3/2} + \frac{1}{8}c^{1/2} - \frac{3}{64}c^{-1/2}. \quad (7.7)$$

These follow by writing the nonlogarithmic term in (6.1) as $-v^{3/2}/4 + v^{1/2}/2 - 3v^{-1/2}/16$ and using $v\partial_v \ell = 1/2$.

In particular,

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}K_n}{\sqrt{n}} = e^{-\gamma/2} \approx 0.749306001288449. \quad (7.8)$$

The decimal is uncertified. Riedel's numerical estimate 0.7719351399 [4, p. 3] is a finite-data estimate of the constant, accompanied there by caution about the fitted exponent. The positive $\frac{c}{2} \log n$ term explains a slow approach to the limit. The exact square-root limit belongs to the pre-existing expansive-assembly regime; the displayed lower-order corrections are specific to these component weights.

Corollary 7.2 (Credited Gaussian regime). *With $\sigma_n^2 = \sqrt{cn}/2$,*

$$\frac{K_n - \mathbb{E}K_n}{\sigma_n} \implies N(0, 1). \quad (7.9)$$

The same statement holds with σ_n^2 replaced by $\text{Var } K_n$.

Proof. Use the analytic logarithm of $f_n(e^t)/f_n(1)$ in a fixed complex neighborhood of $t = 0$. By (7.1) and Cauchy's estimates, its third derivative is $O(\sqrt{n} + \log n)$ there. Its second derivative at 0 is $\sigma_n^2 + O(\log n)$. For fixed real θ , substitute $t = i\theta/\sigma_n$ and subtract $t\mathbb{E}K_n$. Taylor's theorem gives

$$\log \mathbb{E} \exp \left(i\theta \frac{K_n - \mathbb{E}K_n}{\sigma_n} \right) = -\frac{\theta^2}{2} + o(1),$$

with cubic error $O(n^{-1/4})$ and second-derivative replacement error $O(n^{-1/2} \log n)$. The continuity theorem for characteristic functions proves (7.9). Equation (7.3) and Slutsky's theorem justify the variance normalization. $\square$

The proof recovers a known generic limit-law phenomenon with the normalization needed here. It is not a new universality assertion. No local limit theorem or quantitative distribution-distance bound is inferred from this fixed-frequency argument.

Remark 7.3 ([write] Riedel's conjecture, and what his fitted constants missed, 6 October 2026). *The source.* Riedel's note [4] has a paragraph "The conjecture" on p. 3 of 6, the same in the revised version (page footers dated 24.01.26) and in the original linked from A392471 (footers dated 13.01.26). With X the number of components of a uniformly random nested cycle partition on n nodes, and $\mathbb{E}[X]$ computed from his recurrences, the boxed conjecture is

$$\mathbb{E}[X] \sim N\sqrt{n},$$

where the constant N is left to be determined (readers able to contribute are asked to contact him). He adds that the asymptotics might instead involve a power n^M with M "close to but not equal to $1/2$ ", and reports that the numerical data suggest $M \geq 0.4704567259$ and $N \approx 0.7719351399$. The note gives neither the range of n nor the fitting method. Neither OEIS entry contains the conjecture (Section 1.1). His X is K_n : his EGF is $Q(z, u) = \exp(u \sum_{k \geq 1} Q_k(z))$ with $\sum_{k \geq 1} Q_k(z) = P(z) - 1$ (his p. 2), which is $F(z, u)$ of (1.2).

(a) *What is proved.* By (7.8) the boxed conjecture holds, with $N = e^{-\gamma/2} = 0.7493060\dots$; this answers his question for N . The alternative is excluded: for $N > 0$, $\mathbb{E}K_n \sim Nn^M$ holds only for $M = 1/2$, since $\mathbb{E}K_n/n^M = (\mathbb{E}K_n/\sqrt{n})n^{1/2-M}$ tends to $+\infty$ for $M < 1/2$ and to 0 for $M > 1/2$. The inequality $M \geq 0.4704567259$, as printed, is satisfied by $M = 1/2$; the value $N \approx 0.7719351399$ is not the constant: it exceeds $e^{-\gamma/2}$ by 0.0226. The source proves all of this in Theorem 7.1 but calls Riedel's number a finite-data estimate and does not say that the conjecture is settled; it is.

(b) *What the fits missed* [write]. Put $E_n = \mathbb{E}K_n$ and $m_0 = -0.977132\dots$ as in (7.2). Then

$$\frac{E_n}{\sqrt{n}} = e^{-\gamma/2} + \frac{\frac{c}{2} \log n + m_0}{\sqrt{n}} + O\left(\frac{(1 + \log n)^2}{n}\right),$$

and the numerator $\frac{c}{2} \log n + m_0$ is positive exactly when $n > e^{-2m_0/c} = 32.48\dots$; so $E_n/\sqrt{n} > e^{-\gamma/2}$ for all large n and approaches it only at the rate $(\log n)/\sqrt{n}$. For every fixed integer $\varrho \geq 2$ the two-point exponent satisfies

$$M_\varrho(n) := \frac{\log(E_{\varrho n}/E_n)}{\log \varrho} = \frac{1}{2} - (1 - \varrho^{-1/2}) \frac{c \log n}{2\sqrt{cn} \log \varrho} + O(n^{-1/2}),$$

so it lies below $1/2$ for all large n and tends to $1/2$ at the same slow rate. *Proof.* By (7.2), $E_n = \sqrt{cn}(1 + \delta_n)$ with $\delta_n = (\frac{c}{2} \log n + m_0)/\sqrt{cn} + O(n^{-1}(1 + \log n)^2) \rightarrow 0$. Since $\log(1 + x) = x + O(x^2)$, $\log(E_{\varrho n}/E_n) = \frac{1}{2} \log \varrho + \delta_{\varrho n} - \delta_n + O(n^{-1}(1 + \log n)^2)$, and

$$\begin{aligned} \delta_{\varrho n} - \delta_n &= \frac{\varrho^{-1/2}(\frac{c}{2} \log n + \frac{c}{2} \log \varrho + m_0) - (\frac{c}{2} \log n + m_0)}{\sqrt{cn}} + O\left(\frac{(1 + \log n)^2}{n}\right) \\ &= -(1 - \varrho^{-1/2}) \frac{c \log n}{2\sqrt{cn}} + O(n^{-1/2}). \end{aligned}$$

Dividing by $\log \varrho$ gives the claim. $\square$ A fitted exponent below $1/2$ and a fitted constant above $e^{-\gamma/2}$ are therefore what the proved law produces at every large finite n : the additive correction $\frac{c}{2} \log n + m_0$ in (7.2), which is not a power, is read by a fit as a change of exponent and of constant. Values (double-precision evaluation of E_n from recurrences with positive terms, agreeing with the exact rational E_{100} and with a 256-bit fixed-point evaluation at $n = 4500$):

n	100	1000	4500	10000	20000
$E_n/\sqrt{n}$	0.821954	0.786471	0.771935	0.766422	0.762634
$M_2(n)$	0.47781	0.48628	0.49089	0.49285	0.49428

For $1 \leq n \leq 40000$ every computed ratio $E_n/\sqrt{n}$ exceeds $e^{-\gamma/2}$, and the ratios decrease from $n = 4$ on.

(c) Where Riedel's numbers may come from. $E_{4500}/\sqrt{4500} = 0.77193514015$ agrees with his $N \approx 0.7719351399$ to 2.5×10^{-10} , and the ratio first falls below his value at $n = 4501$. This suggests, but the note does not say, that his N is the finite ratio at $n = 4500$. The write could not identify the computation behind $M \geq 0.4704567259$: the one-step log-log slopes $\log(E_n/E_{n-1})/\log(n/(n-1))$ for $n \leq 40000$, the two-point exponents $\log(E_{4500}/E_m)/\log(4500/m)$ for $m < 4500$, and least-squares fits of $\log E_n$ against $\log n$ over the windows $[a, 4500]$, $1 \leq a < 60$, and a few shorter windows were tried, and none reproduces it to its ten digits. Recorded only: nothing was submitted to the OEIS and Riedel was not contacted.

8 Asymptotic inversion with safe integer rounding

For sufficiently large real y , define

$$N(y) = \min\{n \geq 0 : a_n \geq y\}.$$

The sequence is strictly increasing for $n \geq 1$. Adding label $n + 1$ as a new singleton component is an injection; a single cycle on all $n + 1$ labels is outside its image. The equality $a_0 = a_1 = 1$ causes no problem for large y .

Put $\beta = c/2 - 3/4$. For fixed $J \geq 0$, define at sufficiently large real x

$$\begin{aligned} G_J(x) &= \log \Gamma(x + 1) + 2\sqrt{cx} + \beta \log x + \log C \\ &\quad + \log \left(1 + \sum_{j=1}^J x^{-j/2} R_j(\log x; 1) \right). \end{aligned} \tag{8.1}$$

The last logarithm is real and well defined eventually, and $G'_J(x) \sim \log x$. At integer n , Theorem 2.1 implies

$$\log a_n = G_J(n) + O(n^{-q}(\log n)^p), \quad q = (J + 1)/2, \quad p = 2J + 2. \tag{8.2}$$

For large y , let x_J be the unique sufficiently large real solution of $G_J(x_J) = \log y$.

Theorem 8.1 (Two-ceiling enclosure). *For each fixed J there exist $M_J > 0$ and a threshold such that, with*

$$\epsilon_J = M_J x_J^{-q} (\log x_J)^{p-1},$$

one has

$$\lceil x_J - \epsilon_J \rceil \leq N(y) \leq \lceil x_J + \epsilon_J \rceil. \quad (8.3)$$

If $\text{dist}(x_J, \mathbb{Z}) > \epsilon_J$, the two ceilings coincide and $N(y) = \lceil x_J \rceil$. No explicit certified M_J or onset threshold is supplied.

Proof. Bound the integer error in (8.2) by $Kn^{-q}(\log n)^p$. For $x = x_J$ sufficiently large, $\epsilon_J < 1$, $G'_J \geq (\log x)/2$ on $[x-2, x+2]$, and the error at integers in that interval is at most $2Kx^{-q}(\log x)^p$. Choose $M_J > 4K$. Then $n_+ = \lceil x + \epsilon_J \rceil$ satisfies

$$G_J(n_+) - G_J(x) \geq \frac{\log x}{2} \epsilon_J > 2Kx^{-q}(\log x)^p,$$

so $\log a_{n_+} > \log y$. Similarly $n_- = \lceil x - \epsilon_J \rceil - 1 < x - \epsilon_J$ satisfies $\log a_{n_-} < \log y$. Monotonicity gives (8.3). Enlarging the threshold ensures all comparisons used above. $\square$

A single ceiling of a finite continuous asymptotic expansion is not valid unconditionally: a target may lie arbitrarily close to an exact integer threshold. One must use the two-ceiling enclosure, prove the stated separation, or compare exact counts.

For a convenient starting approximation let $L = \log y$ and use the positive real branch of Lambert's W function:

$$x_0^* = \frac{L}{W(L/e)}. \quad (8.4)$$

It solves $x(\log x - 1) = L$, and $N(y) \sim x_0^*$. With $\alpha = c/2 - 1/4$ and $K_0 = \log(C\sqrt{2\pi})$, Stirling's formula gives

$$\begin{aligned} x_* &= x_0^* - \frac{2\sqrt{cx_0^*} + \alpha \log x_0^* + K_0}{\log x_0^*}, \\ x_J &= x_* + O((\log x_0^*)^{-2}). \end{aligned} \quad (8.5)$$

For justification, the displacement from x_0^* is $O(\sqrt{x_0^*}/\log x_0^*)$. Substituting (8.5) into $G_J - L$, the uncanceled derivative correction is $O(1/\log x_0^*)$; the quadratic curvature term is $O((\log x_0^*)^{-2})$ and the omitted Stirling and fixed-order corrections are smaller. Dividing by $G'_J \sim \log x_0^*$ proves the displayed error. Newton iteration or formal reversion of (8.1) gives further continuous orders. Every integer conclusion still needs (8.3) or an exact comparison. The companion's bounded exact inverse compares integer counts directly and does not use an uncertified asymptotic rounding rule.

Remark 8.2 ([write] The inversion against the transseries volume, 6 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`; its letters κ , d , v , X , L are written here with the subscript “vol” where they would collide with this report's.

- (a) *The starting value x_0^* of (8.4) is an instance, verbatim, of `p0:prop:factorial-core` with $(\kappa_{\text{vol}}, d_{\text{vol}}, L_{\text{vol}}) := (1, -1, L)$, $L = \log y > 0$: the phase is $X(\log X - 1)$, the solution is $v_{\text{vol}} = W_0(L/e)$ and $X_{\text{vol}} = e^{v_{\text{vol}}+1} = L/v_{\text{vol}} = x_0^*$ (using $v_{\text{vol}}e^{v_{\text{vol}}} = L/e$), and the side condition $\log X_{\text{vol}} - 1 > 0$ holds because $v_{\text{vol}} > 0$. The “positive real branch” of the source is W_0 .*

- (b) $N(y)$ is the staircase $N_*(y)$ of `p0:def:three-inverses` for the sequence (a_n) with $n_1 = 1$, for every $y > 1$: the sequence is strictly increasing for $n \geq 1$ (the injection argument at the start of this section), and for $y > 1 = a_0 = a_1$ the minima over $n \geq 0$ and over $n \geq 1$ agree.
- (c) *Theorem 8.1 is not an instance of `p0:thm:staircase`*, only an analogue of its item (2). Its proof compares the model G_J with $\log a_n$ at the integers and then uses monotonicity directly; no admissible interpolation of (a_n) appears, and x_J is the root of the model, not the inverse $\nu_{\mathcal{A}}(y)$ of an interpolation, so the hypothesis $|x_* - \nu(y)| \leq \eta$ of item (2) is not established. The conclusion has the two-ceiling form of item (2), and the paragraph after the proof describes the failure mode stated there. The Newton correction (8.5) is proved directly; no comparison with the volume's perturbation series is claimed.

No novelty is claimed for these inversions.

9 Exact recurrences and reproducible checks

The infinite products in the definitions reduce to finite exact computations at every fixed size. The companion uses a deliberately bounded range, not an unbounded enumeration promise.

Let $B_n^{(m)}$ count labeled permutations of n labels having at most one cycle of each length in $\{1, \dots, m\}$. With $B_0^{(0)} = 1$ and all other entries zero,

$$B_n^{(m)} = B_n^{(m-1)} + \begin{cases} \binom{n}{m} (m-1)! B_{n-m}^{(m-1)}, & n \geq m, \\ 0, & n < m. \end{cases} \quad (9.1)$$

The cycle's labels are chosen first and its cyclic order next; using the previous row enforces at most one such cycle. Thus $B_n = B_n^{(n)}$. The component containing label 1 gives the partial Bell recurrence

$$T(n, k) = \sum_{m=1}^n \binom{n-1}{m-1} B_m T(n-m, k-1), \quad n, k \geq 1. \quad (9.2)$$

All arithmetic in these recurrences is integer arithmetic. Their first five nonempty rows and row sums are

n	$T(n, 1), \dots, T(n, n)$	a_n
1	1	1
2	1, 1	2
3	5, 3, 1	9
4	14, 23, 6, 1	44
5	74, 120, 65, 10, 1	270

An independent rational-series check follows from

$$\log P(z) = \sum_{r \geq 1} \ell_r z^r, \quad \ell_r = \sum_{m|r} \frac{(-1)^{r/m+1}}{(r/m) m^{r/m}}.$$

Formal differentiation gives

$$np_n = \sum_{r=1}^n r \ell_r p_{n-r}, \quad nf_n(1) = \sum_{r=1}^n r p_r f_{n-r}(1), \quad p_0 = f_0(1) = 1. \quad (9.3)$$

A separate power-series multiplication checks $T(n, k) = n![z^n](P - 1)^k/k!$ directly. At small sizes, literal permutation cycle decomposition followed by enumeration of admissible groupings of cycles gives an additional combinatorial check.

The frozen OEIS fixtures contain the displayed numeric terms with public source URLs and retrieval metadata: 23 values of A007838, 22 values of A308338, and 55 entries of A392471. They are independent expected values, not values generated during a verification run. The companion checks those prefixes, complete counts and triangle through $n = 100$, independent rational count recurrences through $n = 100$, power-series triangle identities through $n = 30$, and literal enumeration through $n = 7$. Exact rational mean and variance samples come directly from the triangle. Optional SymPy checks verify the finite symbolic correction identities. See the package README for the commands and dependency versions used.

Evidence levels. The fixtures, finite recurrences, exact integer comparisons, rational moments, and symbolic identities are exact. They are finite checks and do not prove an asymptotic theorem. The fixed-order remainder, complex uniformity, all-circle suppression, moment expansion, CLT, and inverse enclosure are proved in the text using the credited component estimate. Their implied constants and onset thresholds have not been made effective here. All decimal constants or other numerical asymptotic diagnostics are uncertified; high arithmetic precision alone would not certify their errors. The package supplies no interval enclosure for the transcendental constant C and no empirical fitted constant as a substitute for its exact definition.

[write] **The package here, and the reruns, 6 October 2026.** “The companion” and “the package README” above are the delivered ones. In this report the companion programs are `code/companion-exact_nested.py`, `code/companion-verify.py`, `code/companion-test_exact_nested.py` and `code/companion-symbolic_checks.py` (delivered in `companion/`); the frozen fixtures and outputs are in `data/`; the companion’s README and provenance note are `companion-README.md` and `companion-PROVENANCE.md`; the release tools `build_pdf.py`, `make_zip.py`, `release_tools.py` and `test_release.py` are in `code/`; `SOURCE_PROVENANCE.json` and `verification_receipt.json` are in `data/`; the checksum manifest is not shipped. `verify.py` reads `../data/published_fixtures.json` relative to its own folder, so the programs run only on a copy with the delivered names (the report README gives two routes). On such copies at the write (Windows, Python 3.14.4, SymPy 1.14.0): from a fresh extraction of the archive, `verify.py`, `verify.py -include-data`, `test_exact_nested.py` and `symbolic_checks.py` reproduce `full_verification.json`, `exact_data_100.json`, `tests.normal.json` and `symbolic.normal.json` once Windows line endings are converted; from the shipped files, the verifier and the regression tests reproduce their outputs the same way. The release tests need POSIX descriptor support and were not run.

10 Source coverage and attribution

The source review covered the three OEIS entries and Riedel’s revised January 2026 note, the 2006 hybrid asymptotics paper, and the later Erlihson–Granovsky expansive-assembly paper. It also checked the connection between the stated component weights and the hypotheses of that general assembly theory. This is bounded source coverage, not a worldwide historical-priority certificate.

The attribution is deliberately separated by mathematical role. Gutkovskiy’s 2019 OEIS entry supplies A308338’s defining EGF. Riedel supplies the nested-cycle interpretation and A392471. Greene and Knuth are credited for (1.4) by the inspected hybrid paper; the original book was not independently inspected. The hybrid paper itself develops the full component expansion, with root-of-unity periodic coefficient terms. Those component results do not by themselves establish the marked outer-exponential expansion proved here.

The inspected Erlihson–Granovsky preprint is the 19 June 2007 arXiv version of the paper

published in October 2008. Theorem and corollary numbers agree with the journal version, while some equation numbers differ. The reference to the 2004 total-component CLT is attribution verified in the later work, not a direct reading of the 2004 paper. That original paper was not inspected for this report.

The variance in (7.3) is derived independently from (7.5). There is a normalization inconsistency in the later source's displayed general- C covariance formula (preprint (4.54), published (4.18)): with its definition $b_r(u) = C\Gamma(r+1, u)$, the printed conditioning denominator is $\Gamma(p+2)$. The hypotheses allow every $C > 0$. At the interior choice $p = 1$, $C = 3$, $u = 1$, literal substitution gives $3/e - 18/e^2 < 0$, which cannot be a variance. This display is therefore not used to certify our constant. The paper remains the cited source for the general regime and its explicit historical attribution.

Within the checked sources, the explicit count prefactor, fixed-order complex-marked outer expansion, logarithmic moment refinements, and rounding-qualified inverse are the sequence-specific development of this report. No claim is made that no other work contains overlapping results. There is no external publication or amendment of an OEIS entry associated with this document.

Remark 10.1 ([write] The covariance display of Erlihson and Granovsky, corrected, 6 October 2026). In this remark $p, C, u, q, N, r_N, \delta, f_j, b_r, \nu, S_j, Z_N$ are the letters of [6]: the exponent and constant in $a_k \sim Ck^{p-1}$ ($k \rightarrow \infty$; every $p > 0$, $C > 0$ is allowed), a scaled component size, the number of strata, the total size, the scaling $r_N = h^{-1}N^{1/(p+1)}$ with $h = (C\Gamma(p+1))^{1/(p+1)}$, $\delta = 1/r_N$, and so on; they are not p_m , the amplitude C , the marking variable u or the q of Section 8.

The display. With $b_r(u) = C\Gamma(r+1, u)$ (preprint (4.53), published (4.17)), Theorem 4.3(i) states that the scaled counts $(\hat{\nu}_1, \dots, \hat{\nu}_q)$ of components of size at least $u_j r_N$ converge to a centred Gaussian vector with covariance

$$e_{mk} = b_{p-1}(u_s) - \frac{b_p(u_m)b_p(u_k)}{\Gamma(p+2)}, \quad s = \max(k, m)$$

(preprint (4.54); published (4.18) on p. 927, identical, read by the write), and Corollary 4.5 ($q = 1$, $u_1 = u$) gives $\hat{\nu}(u) \Rightarrow N(0, e_{11}(u))$ “for all $u \geq 0$ ”, the case $u = 0$ being the total number of components. The scaling of $\hat{\nu}_j$ (preprint (4.45), published (4.9)) multiplies $\nu_j - S_j$ by $(f_0(p-1)/S_0)^{1/2} \sim r_N^{-p/2}$.

(a) *The display fails.* At $p = 1$, $C = 3$ (for instance $a_k = 3$ for all k), $q = 1$, $u_1 = 1$: $b_0(1) = 3\Gamma(1, 1) = 3/e$, $b_1(1) = 3\Gamma(2, 1) = 6/e$ and $\Gamma(3) = 2$, so

$$e_{11} = \frac{3}{e} - \frac{36}{2e^2} = \frac{3}{e} - \frac{18}{e^2} = -1.33240\dots < 0,$$

which is not the variance of any random variable. More generally, at $u = 0$ the display gives $e_{11}(0) = C\Gamma(p) - C^2\Gamma(p+1)^2/\Gamma(p+2) = C\Gamma(p)(1 - Cp/(p+1))$, negative whenever $C > (p+1)/p$. The source found the first value (Section 10).

(b) *The correction.* The denominator should be $C\Gamma(p+2) = b_{p+1}(0)$:

$$e_{mk}^{\text{corr}} = b_{p-1}(u_s) - \frac{b_p(u_m)b_p(u_k)}{C\Gamma(p+2)},$$

which coincides with the printed display exactly when $C = 1$. This is what the conditioning computation gives. In the Boltzmann model at $\delta = 1/r_N$ the counts n_k are independent Poisson variables of means $a_k e^{-\delta k}$; with $\nu_j = \sum_{k \geq u_j r_N} n_k$ and the total mass $Z = \sum_k k n_k$, Riemann sums

give $r_N^{-p} \text{Cov}(\nu_m, \nu_k) \rightarrow C\Gamma(p, u_s) = b_{p-1}(u_s)$, $r_N^{-p-1} \text{Cov}(\nu_m, Z) \rightarrow C\Gamma(p+1, u_m) = b_p(u_m)$ and $r_N^{-p-2} \text{Var } Z \rightarrow C\Gamma(p+2) = b_{p+1}(0)$, and the covariance of a Gaussian vector conditioned on Z is $\text{Cov}(\nu_m, \nu_k) - \text{Cov}(\nu_m, Z) \text{Cov}(\nu_k, Z) / \text{Var } Z$. The corrected display is a variance: $e_{11}^{\text{corr}}(u) = C(\Gamma(p, u) - \Gamma(p+1, u)^2 / \Gamma(p+2)) > 0$, because the Cauchy–Schwarz inequality for $x^{(p-1)/2} e^{-x/2}$ and $x^{(p+1)/2} e^{-x/2}$ on (u, ∞) gives $\Gamma(p+1, u)^2 < \Gamma(p, u) \Gamma(p+2, u) \leq \Gamma(p, u) \Gamma(p+2)$. In the example of (a) it is $3/e - 6/e^2 = 0.29163\dots$. The write checked these limits and the conditioning formula, not the remaining steps of Section 5 of [6].

(c) *Where the factor C is lost.* The published (5.62) on p. 936 (preprint (5.121)) reads $\text{Var } Z_N = \sum_k k^2 a_k e^{-\delta k} \sim \Gamma(p+2) \delta^{-(p+2)}$; under $a_k \sim Ck^{p-1}$ the right side is $C\Gamma(p+2) \delta^{-(p+2)}$. The same $\Gamma(p+2)$ without C enters T_q in (5.69) (preprint (5.128)), while the proof of (5.72) on p. 938 invokes $\sum_j f_j(p+1) = C\Gamma(p+2)$. The covariance (4.13) of Theorem 4.1(ii) (preprint (4.49); published p. 926) carries the same denominator and fails in the same way: at $q=1$, $p=1$, $C=3$, $u_1=10$ its entry $f_0(0) - f_0(1)^2 / \Gamma(3)$ equals $-1.4956\dots$ (corrected: $1.5014\dots$). Their own identity $f_0(p+1) + f_1(p+1) = C\Gamma(p+2)$ and (5.73) give $\det \Theta^*(1) = C T_1$ for their matrix (5.74), so (5.72) as printed holds only for $C=1$: the factor is lost with (5.62) and carried through T_q , while the quadratic form Θ^* , built from the f_j and w_j , carries C correctly. The inversion that produces (4.13) was not checked. (Sentence strengthened after the independent check of 6 October 2026; the first wording read “The write did not trace which step first drops C ”. Writing $a_j = f_j(p+1)/(f_j(p-1)w_j)$ and $b_j = f_j(p)/(f_j(p-1)w_j)$, (5.73) says $a_j = 1/f_j(p-1) + b_j^2 w_j$, and the determinant of (5.74) is $a_0 a_1 (1/w_0 + 1/w_1) - a_1 b_0^2 - a_0 b_1^2 = (f_0(p+1) + f_1(p+1)) / \prod_{j=0}^1 w_j f_j(p-1)$.)

(d) *The display against this report.* The model here has weights $p_m \rightarrow c$ (Section 1), that is $p=1$ and $C=c$ in the letters of [6], with $N=n$ and $r_N = (n/c)^{1/2}$. Read at $u=0$, as their Corollary 4.5 is read there, with the normalization $r_N^{-1/2}$ (the limit of their scaling factor for $u_1 > 0$), the printed display gives $(K_n - S)/r_N^{1/2} \Rightarrow N(0, c(1 - c/2)) = N(0, 0.40384\dots)$ for a deterministic centring S . Corollary 7.2 and (7.3), proved here without their display, give $(K_n - \mathbb{E}K_n)/r_N^{1/2} \Rightarrow N(0, c/2) = N(0, 0.28072\dots)$, because $\text{Var } K_n/r_N \rightarrow c/2$. The two normalized variables differ by a deterministic shift t_n , and if $Y_n \Rightarrow N(0, \alpha)$ and $Y_n + t_n \Rightarrow N(0, \alpha')$ then (t_n) is bounded and each of its limit points t gives $N(t, \alpha) = N(0, \alpha')$, hence $\alpha = \alpha'$. So the printed display contradicts Corollary 7.2 in this model, and the corrected one, $e_{11}^{\text{corr}}(0) = c - c^2/(2c) = c/2$, agrees with it.

The source used the display for nothing and derived (7.3) independently; no report in the repository uses it. Recorded only: nothing was sent to the authors.

11 Further questions

[write] **Added 6 October 2026, batch 108.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question. Questions 1–5 are the source’s, as delivered; what each lacks is what it names. Question 1: explicit constants in (1.4), in the off-saddle bound (4.4) and in Lemma 5.1. Question 2: control of $f_n(u)$ for u beyond a small disc about 1. Question 3: a global analysis of the secondary arcs beyond the all-arc bound (4.1). Question 4: complexity bounds and certified simplification for the calculator. Question 5: a faster exact algorithm with an independent check. The source’s non-claims are collected in Section 1.1. The intake found no false claim in the source and the write found none either; the two corrections concern outside literature (Remarks 7.3 and 10.1). Question 1 would also give the onset in Remark 7.3(b): $\mathbb{E}K_n/\sqrt{n}$ exceeds $e^{-\gamma/2}$ for all large n , and for every computed $n \leq 40000$, but no explicit n_0 is proved. Question 6 is the write’s.

Several extensions require additional work beyond the proved fixed-order statements.

1. Can the component estimates and Gaussian remainders be made effective enough to give certified constants and usable onset thresholds for the count equivalent and inverse enclosure?
2. Can complex marking be controlled beyond a small disc to prove a lattice local limit theorem, moderate deviations, or a quantitative normal-approximation bound for K_n ?
3. How do the periodic component corrections at secondary roots of unity appear in exponentially small outer-coefficient terms? A complete beyond-all-orders expansion would require a separate global analysis and is not implied by the present all-arc bound.
4. Can the fixed-order coefficient calculator be extended to higher corrections with explicit complexity bounds and certified symbolic simplification?
5. Can the exact Bell triangle be computed faster at large n while retaining an independent check and exact integer arithmetic?
6. [write] Remark 10.1 corrects the covariance displays (4.13) and (4.18) of Erlihson and Granovsky [6] by restoring the factor C in the variance of the total mass, and shows that the printed displays fail. Does their proof, with (5.62) and (5.69) corrected, give exactly the corrected displays, and does any other statement of their Sections 4 and 5 depend on the missing factor? What is missing is a line-by-line audit of their Section 5; the write checked only the conditioning computation and the three limits it uses.

These are research questions, not consequences of the finite numerical or symbolic checks.

[write] **Independent check of the write, 6 October 2026.** An adversarial check made by the intake after the write (commit 399df8a49) re-derived the statements marked [write] from their sources. *Remark 7.3.* Riedel’s note was fetched again in both versions (byte-identical to the write’s copies); the paragraph “The conjecture” on p. 3 is paraphrased exactly ($M \geq 0.4704567259$, not $M \approx 0.47$). The check’s own computation (exact rationals for $n \leq 120$, 320-bit fixed point to $n = 4501$, double precision to $n = 40000$; fixed point against exact at $n = 100$: 1.8×10^{-40}) gives $\mathbb{E}K_{4500}/\sqrt{4500} = 0.771935140148$, 2.48×10^{-10} above Riedel’s N ; the ratio is 0.7719368665 at $n = 4499$ and 0.7719334143 at $n = 4501$, the first n below Riedel’s value. The table, $\mathbb{E}K_{100}$, $\text{Var } K_{100}$, the algebra of (b) and the claims “ $> e^{-\gamma/2}$ ” (minimum margin 0.01031, at $n = 40000$) and “strictly decreasing from $n = 4$ ” were confirmed. One further candidate for M also fails: the single-point exponent $\log \mathbb{E}K_n / \log n$ is 0.46923 at $n = 4500$ and is closest to 0.4704567259 at $n = 7510$, off by 6×10^{-8} ; so Riedel’s value is still not identified. *Remark 10.1.* The published pages 925–927 and 936–938 and the preprint numbers were read again; (a) and (b) were re-derived, and the correction was confirmed independently in this very model: exact finite- n variances of the number of components of size at least ur_N (from $\exp(H_S + wH_L)$) give $\text{Var}/r_N = 0.10799, 0.05448, 0.02976$ at $n = 20000$ for $u = 0.5, 1, 2$, against the corrected 0.10817, 0.05458, 0.02971 and the printed 0.21008, 0.12122, 0.05000 (similarly at $n = 5000$; at $u = 0$, 0.28804 against $c/2 = 0.28073$). (5.62), (5.69) and the identity on p. 938 were confirmed on the page images; from (5.73)–(5.74) the check derived $\det \Theta^*(1) = CT_1$, which the intake re-derived symbolically, and (c) now says so. Part (d), its normalization caveat and the shift lemma were confirmed. *Remark 8.2.* The Lambert instance, the strict increase for $n \geq 1$ and the classification as an analogue were confirmed. *The OEIS records*, fetched again with their histories and b-files, are as described, except one attribution: in A308338 revisions #18–#20 are Heinz’s and #21 is the OEIS server installing his b-file (likewise A392471: #38–#41 Heinz, #42 the server). Own exact values of $T(n, k)$ for rows $0 \leq n \leq 40$, of a_n for $n \leq 120$ and of A007838 for $n \leq 40$ equal the b-files. No mathematical error was found. One attribution is corrected, with a dated note keeping the first wording, and the last sentence of Remark 10.1(c) is strengthened, also with a dated note.

[write] **Part II and these questions, 9 October 2026.** Part II (Section 12) answers none of Questions 1–6: its statistic is the largest outer component, not K_n , and it does not reprove or sharpen any statement of Part I. Two of its own questions (Section 23) are the analogues of questions here for that statistic and are recorded beside them: its Question 9 asks at which order the periodic root-of-unity terms of the component

expansion enter the largest-component law, as Question 3 asks for the outer coefficients; its Question 8 asks for effective constants in its estimates, as Question 1 does for the count equivalent and the inverse. Neither is answered.

Part II

Sharp Gumbel errors and optimal continuous calibration for the largest outer component

A research manuscript of 8 October 2026, archive `sharp_gumbel_assemblies.zip`, delivered as `sharp_gumbel_assemblies/article.tex` with two section files and a generated table file (27 pages). Delivered title block “Sharp Gumbel Errors and Optimal Continuous Calibration for Random Assemblies / Largest components of nested-cycle partitions and other assemblies with asymptotically constant weights / Research manuscript prepared for Vladimir Reshetnikov / 8 October 2026”. Printed as delivered after Section 12, which the write added, with `[write]` notes; the manuscript’s Section k is Section $k + 12$ here, its statements and equations $(k.m)$ are $(k+12.m)$, and its Appendices A and B, Tables 1 and 2 and Figures 1 and 2 keep their letters and numbers. Its labels carry the prefix `ncy:gb:`. It uses Part I for the model, its counts and its credits, not as a premise of any proof.

Abstract of this Part (as delivered). For random assemblies with nonnegative component weights $p_k = c + O(k^{-1})$, $c > 0$, we prove a uniform second-order law for the largest component. If τ is the exact size saddle, $L = \log(c/\tau)$, and $G(x) = \exp(-e^{-x})$, the ordinary Gumbel approximation has error

$$d_K(F_n, G) \sim \frac{(2 + \sqrt{5})e^{-(3+\sqrt{5})/2} \log^2 n}{16\sqrt{c}} \frac{1}{\sqrt{n}}.$$

No affine normalization removes this order. An explicit increasing nonlinear calibration attains error $\tau/(2e) + O(\tau/L)$; we prove that $\tau/(2e)$ is the smallest possible leading error among all continuous distribution functions. Retaining the lattice phase improves the error to $O(\tau/L)$. Under $p_k = c + a_*/k + O(\log k/k^2)$, an exponential-integral correction reduces it to $O(\tau^{3/2}L^3)$. The proof compares the conditioned and unconditioned Fourier integrals before approximation, preserving a cancellation needed below the lattice scale. We apply the results to nested-cycle partitions, counted by OEIS A308338, and to labelled forests of paths. The package includes exact integer checks and independent numerical coefficient-ratio calculations.

Research status. The quantitative refinements have complete conventional proofs in this manuscript and were independently checked during preparation. The first-order Gumbel laws and the component coefficient asymptotics are credited to prior work. The literature comparison supports these as proposed new refinements; it does not establish worldwide priority. The manuscript is unrefereed and is not a Lean or Rocq formalization.

12 Provenance, scope and notation of Part II

[Added 9 October 2026.] Part II was added in a second write of this report, after Part I had been written (399df8a49) and independently checked (25df8755d). It is the manuscript *Sharp Gumbel Errors and Optimal Continuous Calibration for Random Assemblies*, which takes the model of Part I and asks a question Part I does not ask: how far the law of the largest outer component is from its Gumbel limit, and how much of that error any normalization can remove. Everything of the manuscript is printed in this Part except its title page (recorded in the source block above), its table of contents and its reference list (merged into this report’s bibliography): its abstract and status paragraph above, its eleven sections as Sections 13–23, and its two appendices, in the manuscript’s order. Its letters are kept (Section 12.5). This section and the notes marked `[write]` are the write’s; all other text of Sections 13–23 and of the appendices is the manuscript’s.

12.1 The manuscript

The archive `sharp_gumbel_assemblies.zip` (987,801 bytes; 24 files in one wrapper directory `sharp_gumbel_assemblies/`, 1,308,260 bytes unpacked) arrived in commit `b28d0850b` (8 October 2026, 08:39 PDT). Its 18 supporting files were placed in this directory by commit `803f4d937` (9 October 2026), which also retired the archive, with the prefix `02-gumbel-`, the Part number; all 18 are byte-identical to the delivery (confirmed at the write). The programs are `code/02-gumbel-*.py` with the delivered `Makefile` as `code/02-gumbel-Makefile`; the data, the generated table file `data/02-gumbel-numerical_tables.tex`, `requirements.txt` and the delivered checksum list `MANIFEST.sha256` are under `data/`; the two figures (PDF, included here, and PNG previews) under `figures/`; the delivered `PROVENANCE.txt` and `VERIFICATION.txt` are `02-gumbel-PROVENANCE.txt` and `02-gumbel-VERIFICATION.txt` at the top of the directory. Not shipped, and retrievable from the arrival commit: the manuscript sources (`article.tex`, `sections_fourier.tex`, `sections_optimal.tex`, `references.bib`), its PDF and the delivery `README.txt`.

The manuscript is `article.tex` (1,168 lines), which inputs `sections_fourier.tex` (356 lines, its Section 3), `sections_optimal.tex` (402 lines, its Sections 6 and 7) and the generated `numerical_tables.tex`, with a BibTeX file of eight entries; a build of the delivered text reproduces its 27-page PDF, with no warnings. Its title page and PDF author field read “Research manuscript prepared for Vladimir Reshetnikov”; its `README` calls it “an AI-assisted, unrefereed manuscript”, and its `PROVENANCE.txt` says it “was prepared in an AI-assisted mathematical research session”; a source comment at the head of `sections_fourier.tex` names an agent, `proveit_scout`, as the author of that section. Its pins (its Appendix B, `02-gumbel-PROVENANCE.txt` and the bibliography entry [8]) are the blob `07022425b718` of Part I’s `article.tex`, retrieved at ProveIt commit `2bac1ea07b98` (7 October 2026, 21:26 PDT), which is Part I’s text before this write, and commit `adc7f1241b42` of the `openai/math` repository, which is outside this repository and was not checked.

How the manuscript was merged. Its 94 labels carry the prefix `ncy:gb`: (92 in its text, 2 in the shipped table file, which is input unchanged through a local wrapper that adds the prefix), and its 102 references to them (69 `\eqref`, 33 `\ref`) were updated. Its two section files are printed at the places where its driver inputs them; the table file and the two figures are input under their prefixed names. Its citation keys are entries of the merged bibliography, marked [Part II], except two: `oeis308338`, which Part I cites too and which is printed once with Part II’s note added, and `FFGPP`, which is Part I’s `hybrid` (its two citations re-keyed, its note added); its key `ProveItNested` is this report at the pin, that is, Part I. Its macros (`\E`, `\Pbb`, `\R`, `\Z`, `\norminf`, `\DK`, `\cC`, `\cE`, `\ph`, and the unused `\N`, `\ee`, `\fracpart`, `\cA`), its `definition` style and the packages `graphicx` and `longtable` were added to the preamble; its `\dd` is Part I’s. Its title page, colours, running heads and table of contents are not printed. No other text was changed. The write labelled the eleven items of Section 23. The numbering, against a build of the delivered text: its Sections 1–11 are Sections 13–23 here, its Theorem 2.2 is Theorem 14.2, Theorem 3.4 is Theorem 15.4, Theorem 4.2 is Theorem 16.2, Proposition 5.1 is Proposition 17.1, Theorem 5.2 is Theorem 17.2, Theorems 6.1 and 7.1 are Theorems 18.1 and 19.1, Theorem 8.1 is Theorem 20.1, Corollary 9.1 is Corollary 21.1; every statement, subsection and equation $k.m$ is $(k+12).m$; Appendices A and B, Tables 1 and 2 and Figures 1 and 2 are unchanged (all 94 labels compared). The manuscript cites nothing of Part I by number.

What the manuscript says about the repository. Its account of Part I (that it “develops marked enumeration asymptotics and fluctuations of the *number* of components”, and, in its `README`, that it “primarily studies enumeration and component-count fluctuations”) is correct. Its proposal to file the package as a separate report `a308338-sharp-largest-components/` was declined at

placement: the collection keeps one report per A-number, and the manuscript treats the same model and sequence, so it is this report's Part II. Its remark on a Potts reconstruction coefficient in `openai/math` (Subsection 13.3) is context, not a premise, and was not checked.

12.2 What Part II answers in Part I

Part I	After Part II
Questions 1–6	None answered. Part II treats the largest outer component M_n , a statistic Part I does not treat (Section 1.1); it does not reprove or sharpen a statement of Part I.
Question 3 (periodic terms in the outer coefficients)	Its analogue for M_n is Part II's Question 9; both open.
Question 1 (effective constants)	Its analogue for Part II's estimates is Part II's Question 8; both open.
The component estimate (1.4)	Part II's only analytic input for nested cycles, quoted as (13.2) from the same source; with $c = a_* = e^{-\gamma}$ it is Part II's refined hypothesis (14.10).
The counts a_n (Section 9)	Part II's A_n (equation (13.1)); its displayed $A_0, \dots, A_{10}$ and its 22 checked terms agree with Part I's data and with A308338.
The limit $e^{-\gamma/2}$ of (7.8)	The same number is $\sqrt{c}$ in (21.1), where it enters the scale $\tau \sim \sqrt{c/n}$ of the largest component, not the mean of K_n .
The contour of Section 4	Not reused. Part II's τ is the exact saddle of the size equation (14.3); Part I's contour radius at $u = 1$ is $\Re s = \sqrt{c/n}$, which is the leading term τ_0 of τ (Proposition 21.2).

Dated notes at the end of Section 1.1 and of Section 11 say the same.

What Part II proves again: nothing of Part I; it rechecks the first 22 counts. *What is new to the report:* everything else. Before the placement no file of the repository cited Panagiotou and Ramzews or Bousquet-Mélou and Weller, or treated the largest component of an expansive assembly.

12.3 What the write read and checked

At placement (the intake dossier, 9 October 2026; own code): the constant κ as the larger of the two stationary values; the corollary constants $e^{\gamma/2}\kappa/16$, $e^{-\gamma/2}/(2e)$, $2e^{-\gamma/2}/e$; the six exact rational distribution values at $n = 50$ by an integer recurrence, equal to the recorded fractions; A308338 $a(0..16)$; and the raw, smooth and lattice columns of `data/02-gumbel-nested_errors.csv` at $n = 1000$ and 10^4 by a per-cutoff tilted coefficient recurrence (no Fourier quadrature), equal to every printed digit at $n = 10^4$ and within $3 \cdot 10^{-10}$ at $n = 1000$, inside the delivery's own recurrence-versus-Fourier difference. It observed that the data do not yet exhibit the asymptotic constants (note at the end of Section 22); no claim was found wrong.

At the write (9 October 2026; Python 3.14.4, NumPy, SciPy, SymPy 1.14.0, mpmath 1.3.0; own code, kept with the intake records and not shipped). Every proof of Sections 15–21 was read line by line, with no error found. Rechecked by hand or with SymPy in particular: the tail polynomials R_1, R_2, R_3 of Lemma 16.1 and the exact geometric tail (16.4) with its expansion; the algebra (16.6), including the skewness term; the skewness sign $-3b\alpha = -B_3S_1/(2V^2)$; the extraction of the L^2 profile (17.1); both stationary values of $|H|$ and their ratio $(2 + \sqrt{5})^2 e^{-\sqrt{5}} = 1.9178\dots$; the second-difference bound (17.5) $(1/G'(-\log 2) = e^2/2, 2/G'(0) = 2e, 1/G'(\log 2) = 2\sqrt{e}$, second difference

$-1/2$); that $G'(x)(h_\tau(x) - x)$ is exactly the smooth polynomial part of U_n ; the derivative (19.4) and its four regions (the minima at $a = 1/4$ are 0.566, 0.8125 and positive, above the stated bounds 0.133, 0.6875 and $1 + (1 - a)(2t + 1)/4$); the E_1 profile and $\|G'\|_\infty = 1/e$ in Theorem 20.1; the saddle algebra of Proposition 21.2; and every decimal constant of the manuscript at 40 digits (one is rounded: note after the proof of Proposition 17.1). Numerically (floating point, not certified), by a third route that is neither the delivery’s Fourier quadrature nor the intake’s per-cutoff recurrence: the whole distribution of M_n for $n = 10^3, 10^4, 10^5$, from the incremental product $\prod_{k \leq m} \exp(a_k z^k) \bmod z^{n+1}$ in which every term is positive; it reproduces all four error columns of `data/02-gumbel-nested_errors.csv` and hence all twelve entries of Table 1, the largest atom against τ/e , the integer quantiles of Corollary 19.2, the explicit saddle, and an uncertified value of E_* (notes at the end of Sections 22 and 23). It reran the two shipped numerical programs on a copy (`nested_numerics.py`, and `forest_numerics.py` with `-recurrence-check`; NumPy 2.4.4, SciPy 1.17.1, 15 and 23 seconds): the six exact fractions are reproduced exactly, and every error column of the three CSV files agrees with the shipped one to relative 2×10^{-8} or better; only the columns that record floating noise of order 10^{-15} (quadrature differences, the saddle residual) differ. *Not read by the write*: Panagiotou–Ramzews, Bousquet–Mélou–Weller (both versions), Mossel–Sly–Sohn and the `openai/math` repository; the manuscript’s accounts of them are reported, not confirmed. Part I’s reading of the hybrid paper [5] is the only reading of it in this report.

12.4 What is not claimed

From the manuscript: the first-order Gumbel limits and the component coefficient asymptotics are prior work (Panagiotou–Ramzews, Theorem 1.2; Bousquet–Mélou–Weller, Proposition 9; the hybrid paper), and the quantitative refinements are “proposed new contributions”; its directed literature audit “does not establish worldwide priority”, and “a broader specialist review remains appropriate”; the manuscript is unrefereed, AI-assisted, checked by agent derivations and audits that “are not human peer review”, and not a Lean or Rocq formalization; the computations check finite arithmetic and numerical coefficient ratios, and the proofs do not depend on them; no directed-rounding interval arithmetic and no certified numerical supremum (the largest path-forest errors are sampled maxima, and node agreement does not bound the omitted Fourier arc); the O -constants are not effective and no finite threshold is supplied, and “moderate sizes can be far from a leading equivalent”; the refined result needs the refined hypothesis (14.10); U_n need not be a distribution function; the E_1 hierarchy is a comparison of fixed scales, not a convergent series; no numerical value of E_* is used; the Potts coefficient met during the research selection is already in Mossel–Sly–Sohn and is not presented as new; no remote repository was changed. The write adds: its checks are floating evaluations (positive terms only, not interval arithmetic) and hand or SymPy computations; its value of E_* is uncertified; nothing was submitted anywhere.

12.5 Reading conventions

Part II keeps its letters. Where they clash with Part I, read them as follows (the write’s own text in Part II uses Part II’s letters).

Symbol	Part II	Part I
p_k, c, a_*	assembly weights with $p_k \rightarrow c$; for nested cycles Part I’s p_m and $c = e^{-\gamma}$; a_* the $1/k$ coefficient, $= c$ for nested cycles	$p_m = B_m/m!, c = e^{-\gamma}$

Symbol	Part II	Part I
a_k, A_n	Poisson means $p_k e^{-\tau k}$; A_n the nested-cycle counts, Part I's a_n ; $A_n^{(\leq m)}$	a_n (A308338)
$B_j, B_k^{\text{comp}}, V$	tilted size moments $\sum k^j a_k$, $B_1(\tau) = n$, $V = B_2$; $B_k^{\text{comp}} = k! p_k$ is Part I's B_m	B_m (A007838); the constants B, B_3 ; $v = cu$
τ, L	the exact size saddle (14.3); $L = \log(c/\tau) \sim \frac{1}{2} \log n$	no τ ; $s = \sqrt{v/n}$; $L = \log n$ or $\log y$
M_n	the largest outer component	$M_J, M_1(L)$; Riedel's M ; the write's $M_\rho(n)$
G, H, h_τ	the Gumbel distribution function; $H = G'(1 - e^{-x})$; the calibration (14.6)	G_J of Section 8; $H_{j,k}, H_j$; the window h
$J_n, U_n, \tilde{U}_n$	calibrated, lattice and refined approximations	J a fixed order; U_j in (3.5)
$R_j(q)$	tail polynomials of Lemma 16.1	$R_j(L; u)$ the correction polynomials of Theorem 2.1
$Q_n(m), Q_n(p), Q_m$	the tail-moment approximation (Theorem 15.4); the integer quantile (Corollary 19.2); $S_1(m) + ma_m$	$Q(X), Q_j$; Riedel's Q
q, β, θ_n	$q = L + x$; $\beta = e^{-x}$; the fractional part; θ also a Fourier variable	$q = (J+1)/2$, $q = Q(\ell_c)$; $\beta = c/2 - 3/4$
δ, κ	$\delta = \tau L^2/(4c)$; $\kappa = (2 + \sqrt{5})e^{-(3+\sqrt{5})/2}$	$\delta = n^{-1/4}$; the loss constants κ, κ'
$E_*, E_1, \mathcal{E}_n$	the affine minimax constant; the exponential integral; a remainder	$E_{n,J}$; the constant E of (3.6); $\mathbb{E}$
$T, T_m, S_j(m), Z_k$	Poisson totals and tails; Poisson variables; Z_n a normalizer	$T(n, k)$ (A392471); $Z(X)$
N, N_k	$\lfloor 1/\tau \rfloor$; multiplicities	$N(y)$; Riedel's N
$C, C(z), d$	generic constants, $ p_k - c \leq C/k$; the path-forest series	$C(u), C$ (the amplitude)
$u, s, \lambda, \alpha, \ell$	a standardized Fourier variable or a probability; $s = \tau m$; $\lambda = S_0(m)$; $\alpha = S_1/\sqrt{V}$ or an affine slope; an index	u the marking variable; s the saddle; $\lambda = \sqrt{vn}$; $\alpha = c/2 - 1/4$; $\ell = \log s$

13 The question and the contribution

A limiting distribution is often much less accurate than its elegant formula suggests. This is particularly visible for the largest component of a random assembly. At a first approximation, rare large components behave like independent Poisson events, producing a Gumbel law. The assembly, however, is sampled with its *total size fixed*. Removing large components changes both the mean and the variance of that total. Although these changes are small on the central-limit scale, their effect on the extreme-value distribution is larger than the spacing of the integer lattice by a factor of order $\log^2 n$.

This manuscript identifies that effect, calculates its exact profile, and removes it. The final continuous approximation is as accurate, to first order, as any continuous approximation can be. We then retain the integer lattice and resolve a further, model-dependent correction.

The concrete motivating sequence is

$$A_n = n! [z^n] \exp(P(z) - 1), \quad P(z) = \prod_{k \geq 1} \left(1 + \frac{z^k}{k}\right). \quad (13.1)$$

These are the nested-cycle partitions of OEIS A308338 [2]. The existing ProveIt report on this sequence develops marked enumeration asymptotics and fluctuations of the *number* of components [8]. Here the random variable is the *size of the largest* outer component. The distinction matters: a largest outer component contains a permutation with distinct cycle lengths; it is not just the longest individual cycle.

13.1 What is already known

The exponential formula and conditioned independent-Poisson representation of assemblies are standard. Panagiotou and Ramzews prove Gumbel convergence for largest clusters in a much broader class of expansive assemblies and multisets [9, Theorem 1.2]. Bousquet-Mélou and Weller previously obtained a Gumbel law for the largest path in a random labelled forest of paths [10, 11, Proposition 9 in the conference version]. Thus a first-order Gumbel limit for our examples is established background, not a new theorem of this manuscript.

For the component series P in (13.1), Flajolet, Fusy, Gourdon, Panario, and Pouyanne record the estimate

$$[z^k]P(z) = e^{-\gamma} + \frac{e^{-\gamma}}{k} + O\left(\frac{\log k}{k^2}\right), \quad (13.2)$$

and develop fuller expansions, including periodic terms [5, Section 3, equation (22), and Proposition 1]. Their estimate is the only non-elementary coefficient input needed for the nested-cycle application. The general probability theorem below is proved from its explicitly stated weight hypothesis.

13.2 The quantitative change

Write M_n for the largest component. The main normalization is $Y_n = \tau M_n - L$, where the exact saddle τ and L are defined in Section 14. In all statements below, $\tau \sim \sqrt{c/n}$ and $L \sim \frac{1}{2} \log n$. The error is uniform in the CDF argument.

Approximation	Error proved here
Ordinary Gumbel G	$\frac{\kappa}{4c} \tau L^2 + O(\tau L)$, with explicit $\kappa > 0$
Best affine normalization	$E_* \frac{\tau L^2}{4c} (1 + o(1))$, with $E_* > 0$
Explicit continuous calibration	$\frac{\tau}{2e} + O(\tau/L)$, with optimal leading constant
Universal lattice approximation	$O(\tau/L)$; a sharp equivalent is obtained when $a_* \neq 0$
Lattice and component correction	$O(\tau^{3/2} L^3)$ under the refined weight hypothesis

The constants in the unqualified O -notation may depend on the fixed weight sequence. They do not depend on x or n . The results are asymptotic estimates, not numerical error certificates valid from a stated finite threshold.

13.3 Relation to the two repositories

The research selection included the four-state Potts reconstruction manuscripts in `openai/math` [12]. A candidate critical cubic coefficient was checked against Mossel–Sly–Sohn [13, Proposition 4.12,

equations (105)–(106)] and found to be already present there. The present problem was selected from the ProveIt sequence program because its error profile and optimal calibration provide a sharper, independently provable target. The probability results below do not depend on a Potts reconstruction claim or on an unverified formal certificate.

[write] **The repositories at the pins, 9 October 2026.** The ProveIt source of this subsection is Part I as written and independently checked: the pinned blob 07022425b718 is Part I's `article.tex` before this write, and the description of Part I in this section is correct. The `openai/math` snapshot and Mossel–Sly–Sohn were not read by the write; the remark on the Potts coefficient is reported, not confirmed, and it is no premise of any proof in this Part.

14 The assembly model and the formulas

Definition 14.1 (Weight hypothesis). Throughout Sections 14–17 and the continuous calibration results, the weights satisfy

$$p_k \geq 0, \quad p_k = c + O(k^{-1}), \quad c > 0. \quad (14.1)$$

Equivalently, after increasing a constant to cover finitely many indices, there is $C < \infty$ such that $|p_k - c| \leq C/k$ for every $k \geq 1$.

For $n \geq 1$, give each vector

$$(N_1, \dots, N_n) \in \mathbb{Z}_{\geq 0}^n, \quad \sum_{k=1}^n k N_k = n,$$

probability

$$\frac{1}{Z_n} \prod_{k=1}^n \frac{p_k^{N_k}}{N_k!}, \quad Z_n = [z^n] \exp \left(\sum_{k \geq 1} p_k z^k \right). \quad (14.2)$$

The law exists for every sufficiently large n , since $p_n > 0$ eventually. Set

$$M_n = \max\{k : N_k > 0\}.$$

If $k!p_k$ is the number of available labelled components of size k , (14.2) is precisely the component spectrum of a uniformly selected labelled assembly.

There is a unique $\tau = \tau_n > 0$ solving

$$B_1(\tau) := \sum_{k \geq 1} k p_k e^{-\tau k} = n. \quad (14.3)$$

The left side is continuous, strictly decreasing from infinity to zero. Define

$$L = \log(c/\tau), \quad Y_n = \tau M_n - L, \quad F_n(x) = \mathbb{P}(Y_n \leq x), \quad G(x) = e^{-e^{-x}}, \quad \delta = \frac{\tau L^2}{4c}. \quad (14.4)$$

Only sufficiently large n , for which $L > 0$, will be used. For each real x , abbreviate

$$\beta = e^{-x}, \quad q = L + x, \quad m = \left\lfloor \frac{q}{\tau} \right\rfloor, \quad \theta_n(x) = \frac{q}{\tau} - \left\lfloor \frac{q}{\tau} \right\rfloor. \quad (14.5)$$

Thus the fractional part is exactly zero at the lattice points $x = \tau m - L$.

The continuous correction is

$$h_\tau(x) = x + \frac{\tau}{4c} \left[q^2 - q - 1 - \beta(q+1)^2 \right], \quad J_n(x) = G(h_\tau(x)). \quad (14.6)$$

The lattice approximation is

$$\begin{aligned} U_n(x) = G(x) + \frac{\tau G(x)}{4c} \left[(\beta - \beta^2)q^2 - (\beta + 2\beta^2)q - \beta - \beta^2 \right] \\ + \tau \left(\frac{1}{2} - \theta_n(x) \right) G'(x). \end{aligned} \quad (14.7)$$

An approximation such as U_n need not itself be a monotone CDF. Its uniform absolute error is the asserted property. In contrast, J_n is a continuous CDF for all sufficiently large n , as proved below.

Theorem 14.2 (Main quantitative conclusions). *Under (14.1), the following hold.*

1. *Uniformly on $\mathbb{R}$,*

$$F_n(x) = G(x) + \delta G(x) e^{-x} (1 - e^{-x}) + O(\tau L). \quad (14.8)$$

Consequently

$$d_K(F_n, G) = \kappa \delta + O(\tau L), \quad \kappa = (2 + \sqrt{5}) e^{-(3+\sqrt{5})/2}.$$

Every affine normalization still has an error of order at least δ , in the precise sense of Theorem 17.2.

2. *Uniformly on $\mathbb{R}$,*

$$F_n(x) = J_n(x) + \tau \left(\frac{1}{2} - \theta_n(x) \right) G'(x) + O(\tau/L). \quad (14.9)$$

In particular

$$d_K(F_n, J_n) = \frac{\tau}{2e} + O(\tau/L).$$

This leading constant is optimal among all continuous CDFs.

3. $\|F_n - U_n\|_\infty = O(\tau/L)$. *If, more precisely,*

$$p_k = c + \frac{a_*}{k} + O\left(\frac{\log(k+1)}{k^2}\right), \quad (14.10)$$

then, with $E_1(v) = \int_v^\infty e^{-u} u^{-1} du$,

$$F_n(x) = U_n(x) - a_* G(x) E_1(\max\{1, L+x\}) + O(\tau^{3/2} L^3) \quad (14.11)$$

uniformly on $\mathbb{R}$. For $a_ \neq 0$,*

$$\|F_n - U_n\|_\infty \sim \frac{|a_*|}{ce} \frac{\tau}{L}.$$

Theorem 15.4 proves the common Fourier estimate. Section 16 derives the universal lattice law. Section 17 treats the ordinary and affine-normalized errors. The optimal continuous and atom results follow in Section 19. Section 20 proves (14.11). These proofs also explain why the successive error scales differ.

14.1 An exact probability representation

Let independent Z_k have Poisson means

$$a_k = p_k e^{-\tau k}, \quad T = \sum_{k \geq 1} k Z_k, \quad T_m = \sum_{1 \leq k \leq m} k Z_k.$$

Their total number is almost surely finite, since $\sum_k a_k < \infty$. Conditional on $T = n$, the exponential factor $\prod e^{-\tau k N_k} = e^{-\tau n}$ is constant, and the law of (Z_k) is (14.2). Define

$$B_j = \sum_{k \geq 1} k^j a_k, \quad V = B_2, \quad S_j(m) = \sum_{k > m, k \geq 1} k^j a_k. \quad (14.12)$$

At the saddle, $B_1 = n$. Independence gives the exact identity

$$\mathbb{P}(M_n \leq m) = e^{-S_0(m)} \frac{\mathbb{P}(T_m = n)}{\mathbb{P}(T = n)}. \quad (14.13)$$

The factor $e^{-S_0(m)}$ is the independent rare-component approximation. The ratio is the correction for conditioning on size.

15 A uniform Fourier estimate for removing the upper tail

This section proves the estimate from which the extreme-value expansions follow. Its assumptions concern only the component weights:

$$p_k \geq 0, \quad p_k = c + O(k^{-1}), \quad c > 0.$$

All constants below may depend on this fixed sequence. For each sufficiently large integer n , let $\tau > 0$ be the solution of

$$B_1(\tau) = n, \quad a_k = p_k e^{-\tau k}, \quad B_j = \sum_{k \geq 1} k^j a_k, \quad V = B_2, \quad L = \log(c/\tau).$$

We suppress the dependence on n and τ . Let Z_k be independent Poisson random variables with means a_k , and write

$$T = \sum_{k \geq 1} k Z_k, \quad T_m = \sum_{1 \leq k \leq m} k Z_k, \quad S_j(m) = \sum_{k > m} k^j a_k.$$

These sums are finite almost surely. Conditional on $T = n$, the vector (Z_k) has precisely the assembly distribution: its weight is proportional to $\prod_k p_k^{Z_k} / Z_k!$, since the factor $e^{-\tau n}$ is constant under the conditioning. In this section M_n denotes its largest occupied component size, and

$$F_n(x) = \mathbb{P}\{\tau M_n - L \leq x\}.$$

The proof uses Fourier inversion on the real circle; no continuation through the circle of convergence is required.

Lemma 15.1 (Cumulants and the saddle). *The saddle exists and is unique. For each fixed integer $j \geq 1$,*

$$B_j = c j! \tau^{-j-1} + O(\tau^{-j}), \quad B_0 = \frac{c}{\tau} + O(\log(1/\tau)).$$

In particular,

$$V = 2c\tau^{-3}(1 + O(\tau)), \quad B_3 = 6c\tau^{-4}(1 + O(\tau)), \quad \tau = \sqrt{c/n} + O(n^{-1}).$$

Proof. The weights are bounded and are at least $c/2$ eventually. Thus B_1 is continuous and strictly decreasing from infinity to zero as τ runs from zero to infinity; its derivative is $-B_2 < 0$. The fixed-order geometric-sum estimates give

$$\sum_{k \geq 1} k^j e^{-\tau k} = j! \tau^{-j-1} + O(\tau^{-j}) \quad (j \geq 1).$$

Multiplying the error $p_k - c = O(k^{-1})$ by $k^j e^{-\tau k}$ adds at most $O(\tau^{-j})$. At order zero the perturbation is bounded by a constant times $\sum e^{-\tau k}/k = O(\log(1/\tau))$. The final assertion follows by substituting $B_1 = n$ into $n = c\tau^{-2} + O(\tau^{-1})$ and solving for τ . Since $p_n > 0$ eventually, $\mathbb{P}(T = n) > 0$ for every sufficiently large n , so the conditional laws used here are well defined. $\square$

Lemma 15.2 (A common block controls the whole circle). *Put $N = \lfloor 1/\tau \rfloor$. There are positive constants d, C such that, for all sufficiently small τ , all $m \geq N$, and $|\theta| \leq \pi$, both characteristic functions satisfy*

$$|\mathbb{E} e^{i\theta T}|, |\mathbb{E} e^{i\theta T_m}| \leq \exp \left[-d \min\{\tau^{-3}\theta^2, \tau^{-1}\} \right].$$

Consequently,

$$\sup_{j \in \mathbb{Z}} \mathbb{P}(T = j) \leq C\tau^{3/2}, \quad \sup_{j \in \mathbb{Z}} \mathbb{P}(T_m = j) \leq C\tau^{3/2} \quad (m \geq N).$$

For either variable, the part of its Fourier integral where $|\theta|\sqrt{V} > L$ is smaller than every fixed power of τ .

Proof. Choose k_0 such that $p_k \geq c/2$ for $k \geq k_0$. On the common block $k_0 \leq k \leq N$, we have $a_k \geq c/(2e)$. We claim that

$$\sum_{k=k_0}^N (1 - \cos k\theta) \geq d_0 \min\{N^3\theta^2, N\} \quad (|\theta| \leq \pi)$$

for a fixed $d_0 > 0$ and all sufficiently large N . If $|\theta| \leq 1/N$, use $1 - \cos u \geq C_0 u^2$ for $|u| \leq 1$ and sum k^2 . If $1/N \leq |\theta| \leq A/N$, where A is a fixed large constant, the same argument restricted to $k \leq \lfloor 1/|\theta| \rfloor$ gives a lower bound $C_1/|\theta| \geq C_1 N/A$. The lower endpoint k_0 only changes the constant when N is sufficiently large. Finally,

$$\left| \sum_{k=k_0}^N e^{ik\theta} \right| \leq \frac{2}{|1 - e^{i\theta}|} \leq \frac{\pi}{|\theta|}.$$

For $|\theta| \geq A/N$, choose, for example, $A = 8\pi$ and subtract this bound from the number of summands. This proves the claim.

The modulus of a Poisson characteristic function is $\exp[-\sum a_k(1 - \cos k\theta)]$. Keeping only the common block proves the asserted bounds for both T and T_m . Fourier inversion then gives the bounds on point probabilities. Since $V \asymp \tau^{-3}$, the integral outside $|\theta|\sqrt{V} \leq L$ is at most

$$C\tau^{3/2}e^{-d_1 L^2} + Ce^{-d_1/\tau}$$

for a possibly smaller $d_1 > 0$. As $L = \log(1/\tau) + O(1)$, this is smaller than every fixed power of τ . $\square$

Lemma 15.3 (The central point probability). *At the saddle,*

$$\mathbb{P}(T = n) = \frac{1}{\sqrt{2\pi V}}(1 + O(\tau)).$$

There is also the following shifted version. If $m \geq N$ and integers ℓ satisfy

$$S_2(m) = o(V), \quad S_1(m) + \ell = o(\sqrt{V}),$$

then

$$\mathbb{P}(T_m = n + \ell) \sim \frac{1}{\sqrt{2\pi V}}.$$

Proof. The centered characteristic function of T is

$$\psi(\theta) = \exp \left\{ \sum_{k \geq 1} a_k (e^{ik\theta} - 1 - ik\theta) \right\}.$$

Put $u = \theta\sqrt{V}$ and $b = B_3/(6V^{3/2}) = O(\sqrt{\tau})$. On $|u| \leq L$, Taylor's theorem and Lemma 15.1 give

$$\psi(u/\sqrt{V}) = e^{-u^2/2} \left[1 - ibu^3 + O(\tau(u^4 + u^6)) \right]. \quad (15.1)$$

Indeed the logarithmic remainder is bounded by $B_4 u^4/(24V^2) = O(\tau u^4)$, and $\sqrt{\tau}L^3 + \tau L^4 \rightarrow 0$, so the exponential Taylor remainder has a uniformly bounded factor. The cubic term integrates to zero. Integrating the remaining error against the Gaussian costs $O(\tau)$; extending the core to the real line and discarding the rest of the original circle is justified by Lemma 15.2. This proves the first formula, including the lower bound $\mathbb{P}(T = n) \geq C^{-1}\tau^{3/2}$.

For the shifted assertion, the variance of T_m is $V - S_2(m)$, and the target $n + \ell$ is displaced from its mean by $S_1(m) + \ell$. On every fixed standardized interval its Fourier integrand therefore converges to $e^{-u^2/2}$. The cubic remainder tends to zero because the third cumulant of T_m is at most B_3 . The common-block bound supplies an integrable Gaussian majorant on the expanding inner arc and a negligible outer arc. Dominated convergence proves the formula. $\square$

Theorem 15.4 (Uniform approximation by tail moments). *For $m \in \mathbb{Z}_{\geq 0}$, define*

$$Q_n(m) = e^{-S_0(m)} \left[1 + \frac{S_2(m) - S_1(m)^2}{2V} - \frac{B_3 S_1(m)}{2V^2} \right].$$

Then

$$\sup_{m \in \mathbb{Z}_{\geq 0}} |\mathbb{P}(M_n \leq m) - Q_n(m)| = O(\tau^{3/2} L^3) = o(\tau).$$

The formula Q_n is an approximation evaluated at the integer cutoff; no assertion that it is a distribution function for every finite n is needed.

Proof. Write $\lambda = S_0(m)$. Independence gives the exact identity

$$\mathbb{P}(M_n \leq m) = e^{-\lambda} \frac{\mathbb{P}(T_m = n)}{\mathbb{P}(T = n)}. \quad (15.2)$$

We first treat a growing central range and then control both complementary tails.

For a real x , put

$$m = \left\lfloor \frac{L+x}{\tau} \right\rfloor, \quad \beta = e^{-x}, \quad s = \tau m.$$

Consider the range

$$\tau^2 \leq \beta \leq 16L. \quad (15.3)$$

Here $L/2 \leq s \leq 4L$ eventually, and hence $m \geq N$. Exponential tail sums and $p_k = c + O(k^{-1})$ imply

$$\lambda = \beta(1 + O(\tau)), \quad S_j(m) \leq C\beta\tau^{-j}L^j \quad (1 \leq j \leq 3). \quad (15.4)$$

For completeness, the constant-weight tail at order zero is $ce^{-\tau(m+1)}/(1 - e^{-\tau}) = \beta(1 + O(\tau))$. The relative perturbation from $p_k - c$ is $O(1/m) = O(\tau/L)$. Higher moments follow by comparing $\sum_{k>m} k^j e^{-\tau k}$ with $\tau^{-j-1} \int_s^\infty u^j e^{-u} du$; the latter is $\tau^{-j-1} e^{-s}$ times a fixed polynomial of degree j . These arguments are uniform throughout (15.3).

The useful cancellation comes from taking the difference of the two Fourier integrals before estimating them. Define

$$D_m(u) = - \sum_{k>m} a_k (e^{iku/\sqrt{V}} - 1), \quad \alpha = \frac{S_1(m)}{\sqrt{V}}, \quad \eta = \frac{S_2(m)}{V}.$$

On $|u| \leq L$,

$$D_m(u) = -i\alpha u + \frac{1}{2}\eta u^2 + R_3(u), \quad |R_3(u)| \leq \frac{S_3(m)}{6V^{3/2}}|u|^3.$$

The bounds above give

$$|\alpha| \leq C\beta\sqrt{\tau}L, \quad \eta \leq C\beta\tau L^2, \quad \frac{S_3(m)}{V^{3/2}} \leq C\beta\tau^{3/2}L^3.$$

In particular $\sup_{|u| \leq L} |D_m(u)| \rightarrow 0$ uniformly in the growing range. A second-order expansion thus gives

$$e^{D_m(u)} - 1 = -i\alpha u + \frac{\eta - \alpha^2}{2}u^2 + R_m(u), \quad (15.5)$$

where

$$|R_m(u)| \leq C\tau^{3/2}L^3(\beta + \beta^2 + \beta^3)(|u|^3 + |u|^6).$$

The terms responsible for this bound are $S_3/V^{3/2}$, $|\alpha|\eta$, and $|\alpha|^3$. The term η^2 is smaller, because $\sqrt{\tau}L \rightarrow 0$. All exponential remainder factors are uniformly bounded.

Exactly,

$$\mathbb{P}(T_m = n) - \mathbb{P}(T = n) = \frac{1}{2\pi\sqrt{V}} \int_{-\pi\sqrt{V}}^{\pi\sqrt{V}} \psi(u/\sqrt{V})(e^{D_m(u)} - 1) du.$$

On the core, multiply (15.5) by (15.1). The linear and cubic terms are odd and integrate to zero. The quadratic term contributes $(\eta - \alpha^2)/2$. The cubic-linear product is

$$(-ibu^3)(-i\alpha u) = -b\alpha u^4,$$

so its Gaussian integral contributes

$$-3b\alpha = -\frac{B_3 S_1(m)}{2V^2}.$$

This determines the negative sign of the skewness term. Every remaining product has Gaussian integral at most

$$C\tau^{3/2} [1 + L^3(\beta + \beta^2 + \beta^3)].$$

In particular the $O(\tau(u^4 + u^6))$ error in the full characteristic function is multiplied by $e^{D_m} - 1$, whose leading coefficient is $O(\beta\sqrt{\tau}L)$; there is no uncanceled $O(\tau)$ constant error. This is why separate estimates of the two probabilities would lose the stated precision.

Outside the core one must bound the original characteristic functions:

$$|\psi(e^{D_m} - 1)| \leq |\psi e^{D_m}| + |\psi|.$$

The two terms are the moduli of the characteristic functions of T_m and T , respectively. Lemma 15.2 therefore makes their integrals negligible uniformly; no separate bound on e^{D_m} outside the core is used. We have proved

$$\begin{aligned} \sqrt{2\pi V} [\mathbb{P}(T_m = n) - \mathbb{P}(T = n)] &= \frac{S_2(m) - S_1(m)^2}{2V} - \frac{B_3 S_1(m)}{2V^2} \\ &\quad + O(\tau^{3/2}[1 + L^3(\beta + \beta^2 + \beta^3)]). \end{aligned}$$

Divide by Lemma 15.3 and use (15.2). Multiplication by $e^{-\lambda}$ bounds every fixed power of β , by (15.4). The extra product from division by $1 + O(\tau)$ is $O(\tau^2 L^2(\beta + \beta^2))$ before exponential weighting, and is absorbed into $O(\tau^{3/2} L^3)$. This proves the theorem in the central range.

It remains to check that this range did not exclude a larger error. Every integer cutoff can be represented by $x = \tau m - L$, so the following two bounds complete the supremum over all m .

If $\beta < \tau^2$, the Poisson identity

$$\mathbb{E}[Z_k \mid T = n] = a_k \frac{\mathbb{P}(T = n - k)}{\mathbb{P}(T = n)} \leq C a_k$$

follows from Lemmas 15.2 and 15.3; probabilities at negative integers are zero. Consequently $\mathbb{P}(M_n > m) \leq C S_0(m) \leq C\beta = O(\tau^2)$. Here $s \geq 3L + O(1)$, and bounded weights give

$$S_j(m) \leq C\beta\tau^{-j}(1 + s)^j \quad (j = 0, 1, 2).$$

The three moment corrections in Q_n are bounded by constants times $\tau\beta(1 + s)^2$, $\tau\beta^2(1 + s)^2$, and $\tau\beta(1 + s)$. Since $\beta = (c/\tau)e^{-s}$, their maxima on this tail are $O(\tau^3 L^2)$. Thus $Q_n(m) = 1 + O(\tau^2)$, which is more accurate than required.

If $\beta > 16L$, put

$$m_0 = \left\lfloor \frac{L - \log(16L)}{\tau} \right\rfloor.$$

Eventually $m_0 \geq N$, $m \leq m_0$, and $S_0(m_0) \geq 8L$. Conditioning and the point-probability bounds give

$$\mathbb{P}(M_n \leq m) \leq \mathbb{P}(M_n \leq m_0) \leq C e^{-S_0(m_0)} = O(\tau^8).$$

For Q_n , use $S_j(m) \leq B_j$ and Lemma 15.1 to bound the absolute value of its bracket by $C\tau^{-1}$. Hence $|Q_n(m)| \leq C\tau^{-1}e^{-8L} = O(\tau^7)$, including cutoffs below N . Both errors are negligible. The central estimate and the two tails prove the asserted uniform bound. Finally $\tau^{3/2}L^3 = o(\tau)$ because $\sqrt{\tau}L^3 \rightarrow 0$. $\square$

16 The universal lattice expansion

We now extract the explicit formulas from the tail-moment estimate. The argument keeps the fractional part of the cutoff. At precision $o(\tau)$, replacing an integer cutoff by a real one without this term changes the answer.

Lemma 16.1 (Geometric tail moments). *In the growing central range*

$$\tau^2 \leq \beta = e^{-x} \leq 16L, \quad (16.1)$$

one has $q = L + x \asymp L$ and, for $j = 1, 2, 3$,

$$S_j(m) = \beta \tau^{-j} R_j(q) + O\left(\tau \beta \tau^{-j} (1+q)^j\right), \quad (16.2)$$

where

$$R_1(q) = q + 1, \quad R_2(q) = q^2 + 2q + 2, \quad R_3(q) = q^3 + 3q^2 + 6q + 6.$$

In addition,

$$S_0(m) = \beta + \tau \beta \left(\theta_n(x) - \frac{1}{2} \right) + O\left(\frac{\tau \beta}{L} + \tau^2 \beta \right). \quad (16.3)$$

All estimates are uniform in (16.1).

Proof. For small τ , (16.1) gives $L - \log(16L) \leq q \leq L + 2\log(1/\tau)$, so $L/2 \leq q \leq 4L$. The constant-weight tail is an elementary exponential sum. Comparing it with the integral gives

$$c \sum_{k>m} k^j e^{-\tau k} = c \tau^{-j-1} \Gamma(j+1, \tau m) + O\left(e^{-\tau m} (1+m)^j\right).$$

For an integer $j \geq 0$, $\Gamma(j+1, s) = e^{-s} j! \sum_{\ell=0}^j s^\ell / \ell!$. The identity $\tau m = q - \tau \theta_n(x)$, together with $ce^{-q}/\tau = \beta$, gives (16.2). The perturbation $|p_k - c| \leq C/k$ contributes at most the corresponding tail with one lower power of k ; its relative size here is $O(\tau/L)$, which is within the displayed error.

For S_0 the constant-weight sum is exact:

$$c \sum_{k>m} e^{-\tau k} = \beta \frac{\tau}{e^\tau - 1} e^{\tau \theta_n(x)}. \quad (16.4)$$

Since $0 \leq \theta_n(x) < 1$,

$$\frac{\tau}{e^\tau - 1} e^{\tau \theta_n(x)} = 1 + \tau \left(\theta_n(x) - \frac{1}{2} \right) + O(\tau^2)$$

uniformly. The weight perturbation is at most $C \sum_{k>m} e^{-\tau k}/k = O(e^{-q}/q) = O(\tau \beta/L)$. This proves (16.3). $\square$

Theorem 16.2 (Uniform lattice law). *Under (14.1), the approximation U_n in (14.7) satisfies*

$$\|F_n - U_n\|_\infty = O(\tau/L). \quad (16.5)$$

Proof. Apply Theorem 15.4 and Lemma 16.1. Lemma 15.1 gives $V = 2c\tau^{-3}(1 + O(\tau))$ and $B_3 = 6c\tau^{-4}(1 + O(\tau))$. Consequently, in (16.1),

$$\begin{aligned} \frac{S_2 - S_1^2}{2V} - \frac{B_3 S_1}{2V^2} &= \frac{\tau}{4c} \left[\beta(q^2 + 2q + 2) - \beta^2(q+1)^2 - 3\beta(q+1) \right] + \mathcal{E}_n(x) \\ &= \frac{\tau}{4c} \left[(\beta - \beta^2)q^2 - (\beta + 2\beta^2)q - \beta - \beta^2 \right] + \mathcal{E}_n(x), \end{aligned} \quad (16.6)$$

where $e^{-\beta/2}|\mathcal{E}_n(x)| = O(\tau^2 L^2)$. The coefficient $-3\beta(q+1)$ comes from the skewness term. Omitting it would leave an error of order τL .

Exponentiating (16.3) yields

$$e^{-S_0} = G(x) + \tau \left(\frac{1}{2} - \theta_n(x) \right) G(x)\beta + O(\tau/L)$$

uniformly in the central range. To see the uniformity directly, the perturbation of S_0 is $o(1)$ there, and the remainders are bounded by fixed multiples of

$$e^{-\beta/2} \left(\frac{\tau\beta}{L} + \tau^2\beta + \tau^2\beta^2 \right).$$

Each polynomial in β times $e^{-\beta/2}$ is bounded. Products with (16.6) cost $O(\tau^2 L^2)$. The Fourier remainder is $O(\tau^{3/2} L^3)$. Both are $o(\tau/L)$. This proves (16.5) on the central range.

For completeness, the extension is uniform on the entire real line. If $\beta < \tau^2$, the Poisson single-component identity and the local concentration bounds in Section 15 give

$$1 - F_n(x) \leq C \sum_{k>m} a_k \leq C\beta.$$

The upper tails of G and the displayed corrections in U_n are $O(\tau^2)$ in this range. If $\beta > 16L$, use the cutoff at $\beta = 16L$. Formula (14.13) and the common local upper and lower bounds give probability $O(e^{-8L})$ there. Monotonicity transfers that bound farther left. Every explicit term of U_n is bounded by a polynomial in $L, \beta, |\log \beta|$ times $e^{-\beta}$, hence is also negligible on this tail. This includes negative integer cutoffs. $\square$

17 The sharp ordinary error and the affine obstruction

Define

$$H(x) = G'(x)(1 - e^{-x}) = G(x)e^{-x}(1 - e^{-x}).$$

Since $q = L - \log \beta$, extracting the L^2 term from (14.7) gives

$$F_n(x) = G(x) + \delta H(x) + O(\tau L) \tag{17.1}$$

uniformly. Indeed, for every fixed nonnegative integer r and positive integer j ,

$$\sup_{\beta>0} e^{-\beta} \beta^j |\log \beta|^r < \infty.$$

All remaining polynomial terms are therefore $O(\tau L)$; the fractional-part term is $O(\tau)$.

Proposition 17.1 (Exact leading Kolmogorov constant). *Let*

$$\beta_+ = \frac{3 + \sqrt{5}}{2}, \quad \kappa = (2 + \sqrt{5})e^{-\beta_+} = 0.309004785987676 \dots$$

Then

$$d_K(F_n, G) = \kappa\delta + O(\tau L) \sim \frac{\kappa}{16\sqrt{c}} \frac{\log^2 n}{\sqrt{n}}. \tag{17.2}$$

The same leading equivalent holds for the elementary normalization $\sqrt{c/n} M_n - \frac{1}{2} \log(cn)$.

Proof. The triangle inequality in the supremum norm applied to (17.1) gives $d_K(F_n, G) = \delta \|H\|_\infty + O(\tau L)$. For $\beta > 0$, $|H| = e^{-\beta} \beta |1 - \beta|$. Its two interior stationary points are $\beta_\pm = (3 \pm \sqrt{5})/2$. Their values are $(\sqrt{5} - 2)e^{-\beta_-}$ and $(2 + \sqrt{5})e^{-\beta_+}$, respectively, and the latter is larger. For example their ratio is $(2 + \sqrt{5})^2 e^{-\sqrt{5}} > 1$; this follows from $\sqrt{5} < 9/4$, $e < 3$, and $3^{9/4} < 16 < (2 + \sqrt{5})^2$. The boundary values are zero. Hence $\|H\|_\infty = \kappa$.

The cumulant estimate gives $\tau = \tau_0 + O(\tau_0^2)$, $\tau_0 = \sqrt{c/n}$. If $L_0 = \log(c/\tau_0)$, then

$$\tau M_n - L = a(\tau_0 M_n - L_0) + b, \quad a = \tau/\tau_0 = 1 + O(\tau_0), \quad b = (a - 1)L_0 + \log a = O(\tau_0 L_0).$$

The functions G', xG', H', xH' are bounded, so this change of normalization changes (17.1) by $O(\tau_0 L_0)$. Finally $\delta \sim \log^2 n / (16\sqrt{cn})$. $\square$

[write] **A rounded constant, 9 October 2026.** The decimal printed for κ above, 0.309004785987676..., is rounded in its last place: $\kappa = 0.3090047859876757102...$ (the write's 40-digit evaluation), so its truncation to the printed length is 0.309004785987675.... The other stationary value is $(\sqrt{5} - 2)e^{-\beta_-} = 0.161120703062022...$. Nothing in this Part depends on the last digit.

The error profile changes sign at $x = 0$. Its ratio to G' is $1 - e^{-x}$, which is nonlinear. That observation is the obstruction to eliminating the leading error by shifting and scaling.

Theorem 17.2 (Best affine normalization). *Define the finite positive constant*

$$E_* = \inf_{\alpha, b \in \mathbb{R}} \sup_{x \in \mathbb{R}} G'(x) |1 - e^{-x} - \alpha x - b|. \quad (17.3)$$

Then

$$\inf_{a > 0, b \in \mathbb{R}} d_K(\mathcal{L}(aY_n + b), G) = E_* \delta + o(\delta). \quad (17.4)$$

Here $d_K(\mathcal{L}(X), G)$ denotes the distance between the CDF of X and G . In particular no sequences of centers and positive scales yield $o(\log^2 n / \sqrt{n})$ error. A concrete bound is

$$\frac{1}{2(e^2/2 + 2e + 2\sqrt{e})} \leq E_* \leq \kappa. \quad (17.5)$$

Proof. For fixed α, b , set $a_n = 1 + \delta\alpha$ and $b_n = \delta b$. Uniform Taylor expansion of (17.1) at $(x - b_n)/a_n$ gives

$$F_n((x - b_n)/a_n) - G(x) = \delta \{H(x) - (\alpha x + b)G'(x)\} + o(\delta).$$

Uniformity follows from bounded weighted first and second derivatives of G and H . Taking the infimum yields the upper bound in (17.4).

For the reverse inequality, take any near-minimizing affine parameters. The identity choice already has error $O(\delta)$. Thus

$$\|G((\cdot - b_n)/a_n) - G\|_\infty = O(\delta).$$

Evaluating at $x = 0, 1$, strict monotonicity of G first gives $b_n \rightarrow 0$, $a_n \rightarrow 1$, and then the local positive lower bound on G' gives $b_n = O(\delta)$, $a_n - 1 = O(\delta)$. Every subsequence has a further subsequence on which $(a_n - 1)/\delta$ and b_n/δ converge. Divide the pointwise expansion by δ along that subsequence and take a supremum to obtain at least E_* . This proves the lower bound.

For positivity and the explicit estimate, put $r = \log 2$ and $f(x) = 1 - e^{-x}$. Its second difference at $-r, 0, r$ is

$$f(-r) - 2f(0) + f(r) = -\frac{1}{2}.$$

An affine function has zero second difference. If the weighted uniform error is E , the triangle inequality gives

$$\frac{1}{2} \leq E \left(\frac{1}{G'(-r)} + \frac{2}{G'(0)} + \frac{1}{G'(r)} \right) = E(e^2/2 + 2e + 2\sqrt{e}).$$

This proves the lower bound. Choosing the zero affine function gives $E_* \leq \|H\|_\infty = \kappa$. The objective in (17.3) is coercive in its two parameters, as is already seen by evaluating at 0 and 1; thus the infimum is attained. No numerical solution of this auxiliary minimax problem is used. $\square$

18 Largest atoms and the obstruction to continuous approximation

The lattice spacing of $Y_n = \tau M_n - L$ is τ . To determine the accuracy attainable by a continuous approximation, we need its largest atom. An $O(\tau)$ approximation to the distribution function cannot be differenced to obtain this information. Instead, we condition directly on the number of components of one specified size.

Theorem 18.1 (Largest atom and continuous approximation). *Under the standing assumptions,*

$$\max_{1 \leq m \leq n} \mathbb{P}(M_n = m) = \frac{\tau}{e} + O(\tau^2 L^2). \quad (18.1)$$

Let $\mathcal{C}$ be the set of continuous distribution functions on $\mathbb{R}$. Then

$$\inf_{J \in \mathcal{C}} \|F_n - J\|_\infty = \frac{1}{2} \max_m \mathbb{P}(M_n = m) = \frac{\tau}{2e} + O(\tau^2 L^2). \quad (18.2)$$

Proof. Write $a_k = p_k e^{-\tau k}$ for the Poisson means. For an integer m , put

$$\beta_m = \frac{c}{\tau} e^{-\tau m}, \quad R_m = \sum_{k < m} k Z_k, \quad Q_m = S_1(m) + m a_m, \quad W_m = V - S_2(m) - m^2 a_m.$$

Thus $\mathbb{E}R_m = n - Q_m$ and $\text{Var}(R_m) = W_m$. Independence, followed by conditioning on Z_m , gives the exact identity

$$\mathbb{P}(M_n = m) = e^{-S_0(m) - a_m} \sum_{\ell \geq 1} \frac{a_m^\ell}{\ell!} \frac{\mathbb{P}(R_m = n - \ell m)}{\mathbb{P}(T = n)}. \quad (18.3)$$

Probabilities on the right are under the independent Poisson law; a negative target for R_m has probability zero.

Fix a sufficiently large constant B , for example $B = 16$, and first consider the integer window

$$\tau^2 \leq \beta_m \leq BL. \quad (18.4)$$

Here $\tau m \asymp L$, uniformly, and $m - 1$ eventually exceeds $\lfloor 1/\tau \rfloor$. The uniform concentration estimate of Lemma 15.2 and the full-law local lower bound of Lemma 15.3 imply

$$\sup_{\ell \geq 1} \frac{\mathbb{P}(R_m = n - \ell m)}{\mathbb{P}(T = n)} \leq C. \quad (18.5)$$

The tail estimates give, uniformly in this window,

$$\begin{aligned} a_m &= \tau \beta_m (1 + O(\tau/L)), & S_0(m) &= \beta_m (1 + O(\tau)), \\ Q_m &= O(\beta_m L/\tau), & \frac{V - W_m}{V} &= O(\tau \beta_m L^2). \end{aligned} \quad (18.6)$$

In particular, $a_m = o(1)$ and $W_m/V \rightarrow 1$ uniformly.

The terms $\ell \geq 2$ in (18.3) are bounded, using (18.5), by

$$Ce^{-S_0(m)}e^{-a_m}(e^{a_m} - 1 - a_m) \leq Ce^{-S_0(m)}a_m^2 \leq C\tau^2\beta_m^2e^{-\beta_m} = O(\tau^2).$$

The constant-factor comparison between $e^{-S_0(m)}$ and $e^{-\beta_m}$ is uniform because $\tau\beta_m = O(\tau L) = o(1)$.

We record the quantitative local estimate needed for the term $\ell = 1$. Put $C_{3,m} = \sum_{k \leq m} k^3 a_k$ and $\phi(u) = (2\pi)^{-1/2}e^{-u^2/2}$. Uniformly over integers j and truncations in (18.4),

$$\begin{aligned} \mathbb{P}(R_m = j) &= \frac{\phi(u)}{\sqrt{W_m}} \left[1 + \frac{C_{3,m}}{6W_m^{3/2}}(u^3 - 3u) \right] + O\left(\frac{\tau}{\sqrt{W_m}}\right), \\ u &= \frac{j - \mathbb{E}R_m}{\sqrt{W_m}}. \end{aligned} \tag{18.7}$$

To justify the uniform remainder, apply the characteristic-function expansion used in Lemma 15.3 to R_m , with its own variance. For $|t| \leq L$, it reads

$$\mathbb{E} \exp\left(\frac{it(R_m - \mathbb{E}R_m)}{\sqrt{W_m}}\right) = e^{-t^2/2} \left[1 - \frac{iC_{3,m}t^3}{6W_m^{3/2}} + O(\tau(t^4 + t^6)) \right].$$

Here $W_m \asymp \tau^{-3}$, the standardized third cumulant is $O(\sqrt{\tau})$, and the fourth-order logarithmic remainder is $O(\tau t^4)$, all uniformly in m . The common Fourier block from Lemma 15.2 bounds the omitted arc by a quantity smaller than every fixed power of τ . Fourier inversion of the cubic term gives $(u^3 - 3u)\phi(u)$; the absolute integral of the remaining Gaussian error is $O(\tau)$. The external phase has modulus one for every j , which proves the stated uniformity in the target integer.

For $j = n - m$, the standardized displacement in this estimate is

$$u_m = \frac{Q_m - m}{\sqrt{W_m}} = O(\sqrt{\tau}L(1 + \beta_m)) = o(1)$$

uniformly in (18.4). The variance factor differs from one by $O(\tau\beta_m L^2)$; the Gaussian factor differs from one by $O(u_m^2)$; and the first Edgeworth term is $O(\sqrt{\tau}|u_m|)$, since its Hermite polynomial is $u^3 - 3u$. The normalized local remainder is $O(\tau)$. The full-law denominator has normalized value $1 + O(\tau)$. Consequently,

$$\frac{\mathbb{P}(R_m = n - m)}{\mathbb{P}(T = n)} = 1 + O(\tau L^2(1 + \beta_m)^2). \tag{18.8}$$

Inserting (18.6) and (18.8) into (18.3) yields

$$\mathbb{P}(M_n = m) = \tau\beta_m e^{-\beta_m} + O(\tau^2 L^2), \tag{18.9}$$

uniformly in the window. Indeed, every error involving β_m is multiplied by an exponentially decreasing factor, and $\sup_{\beta > 0} e^{-\beta}\beta(1 + \beta)^2 < \infty$.

The omitted atoms cannot change the maximum at this precision. If $\beta_m < \tau^2$, the same uniform local upper bound in (18.3) gives $\mathbb{P}(M_n = m) \leq Ca_m = O(\tau^3)$. For the other tail, let

$$m_- = \left\lceil \frac{L - \log(BL)}{\tau} \right\rceil.$$

The exact conditioning identity for $\mathbb{P}(M_n \leq m_-)$, together with Lemmas 15.2 and 15.3, gives

$$\mathbb{P}(M_n \leq m_-) \leq C e^{-S_0(m_-)} \leq C e^{-BL/2} = O(\tau^{B/2}).$$

Every m with $\beta_m > BL$ lies below this threshold.

The function $\beta e^{-\beta}$ has maximum $1/e$, at $\beta = 1$. At $m = \lfloor L/\tau \rfloor$, we have $\beta_m = e^{O(\tau)}$ and $\beta_m e^{-\beta_m} = e^{-1} + O(\tau^2)$. Equation (18.9) now proves (18.1).

For completeness, the first equality in (18.2) is an elementary fact about any finitely supported distribution. A jump of size w forces a continuous approximant to have error at least $w/2$: its value at the jump must be within the claimed error of both one-sided limits. Conversely, list the support as $y_1 < \dots < y_r$, with masses w_i and cumulative sums s_i . Assign a continuous interpolant the values

$$J(y_i) = \frac{s_{i-1} + s_i}{2}, \quad s_0 = 0,$$

and interpolate linearly between consecutive support points. Connect monotonically to zero before y_1 and to one after y_r . On (y_i, y_{i+1}) , the original distribution function equals s_i , whereas the interpolant ranges from $s_i - w_i/2$ to $s_i + w_{i+1}/2$. Its error is therefore at most $\frac{1}{2} \max_i w_i$, everywhere. This proves the equality and completes the theorem. $\square$

An increasing bijection preserves every atom size. Thus the lower bound in (18.2) also applies after any such normalization of the largest component. The comparison class here is the class of continuous distribution functions. A formula retaining the lattice phase, such as Theorem 16.2, can have a smaller remainder because it records the jumps explicitly.

19 An explicit optimal continuous calibration

Retaining the constant terms in the smooth correction produces a continuous approximation with the best possible leading error constant. Define

$$\begin{aligned} q &= L + x, & \beta &= e^{-x}, \\ h_\tau(x) &= x + \frac{\tau}{4c} \left[q^2 - q - 1 - \beta(q+1)^2 \right], & J_n(x) &= G(h_\tau(x)). \end{aligned} \tag{19.1}$$

Theorem 19.1 (Asymptotically optimal smooth calibration). *For $\tau/c \leq 1/4$, the map h_τ is an increasing bijection of $\mathbb{R}$, and J_n is a continuous distribution function. Uniformly for $x \in \mathbb{R}$,*

$$F_n(x) = J_n(x) + \tau \left(\frac{1}{2} - \left\{ \frac{L+x}{\tau} \right\} \right) G'(x) + O(\tau/L). \tag{19.2}$$

Consequently,

$$\|F_n - J_n\|_\infty = \frac{\tau}{2e} + O(\tau/L). \tag{19.3}$$

Equivalently,

$$\sup_{z \in \mathbb{R}} |\mathbb{P}(h_\tau(Y_n) \leq z) - G(z)| = \frac{\tau}{2e} + O(\tau/L).$$

Together with (18.2), this shows that the explicit calibration attains the smallest possible leading constant among continuous approximations.

Proof. First we check monotonicity globally. Put $a = \tau/c$ and use $e^{-x} = (c/\tau)e^{-q}$. Direct differentiation gives

$$(h_\tau)'(x) = 1 + \frac{a}{4}(2q - 1) + \frac{1}{4}e^{-q}(q^2 - 1). \quad (19.4)$$

For $q \geq 1$, both additional terms are nonnegative. For $-1 \leq q \leq 1$, the right side is at least $1 - 3a/4 - e/4 > 0$ when $a \leq 1/4$. For $-2 \leq q \leq -1$, it is at least $1 - 5a/4 > 0$. Finally, for $q \leq -2$, put $t = -q \geq 2$. The inequality $e^t(t^2 - 1) \geq 2t + 1$ follows by evaluating the difference at two and differentiating it on $[2, \infty)$. Thus (19.4) is at least $1 + (1 - a)(2t + 1)/4 > 0$ in this last region as well. The limits of $h_\tau(x)$ at $-\infty$ and $+\infty$ are $-\infty$ and $+\infty$, respectively. This proves the first claim.

Theorem 16.2 gives the uniform lattice expansion with smooth correction $G'(x)(h_\tau(x) - x)$. On the central window $\tau^2 \leq \beta \leq 16L$, we have $q \asymp L$ and

$$|h_\tau(x) - x| \leq C\tau L^2(1 + \beta) = o(1).$$

Taylor's theorem and the exponential decrease of the Gumbel derivatives give, uniformly on this window,

$$G(h_\tau(x)) = G(x) + G'(x)(h_\tau(x) - x) + O(\tau^2 L^4).$$

For example, the second-order remainder is bounded by a constant times $\tau^2 L^4 e^{-\beta/2} \beta(1 + \beta)^3$, whose supremum is finite.

Here are explicit controls for the complementary tails. At the right endpoint $x_+ = -2 \log \tau$, the correction $h_\tau(x_+) - x_+$ is eventually positive, and $1 - G(x_+) \leq \tau^2$. Monotonicity therefore gives $1 - J_n(x) \leq \tau^2$ for $x \geq x_+$. On that interval,

$$|G'(x)(h_\tau(x) - x)| \leq C\tau e^{-x}(1 + (L + x)^2) \leq C\tau^3 L^2.$$

The last inequality follows by differentiating the exponentially weighted polynomial, which is decreasing there. Thus the Taylor replacement costs $O(\tau^2)$ on the right tail.

At the left endpoint $x_- = -\log(16L)$, the correction is $O(\tau L^3) = o(1)$. Hence $J_n(x_-) \leq e^{-8L}$ for sufficiently small τ , and monotonicity extends this bound to $x \leq x_-$. On the same interval $G(x) \leq e^{-16L}$, while

$$|G'(x)(h_\tau(x) - x)| \leq C\tau e^{-\beta} \beta(1 + \beta)(1 + (L - \log \beta)^2) \leq C\tau(1 + L^2)e^{-8L}.$$

The polynomial and logarithmic factors in the middle expression are absorbed by half of its exponential factor. This proves the required replacement on the left tail as well. Since $\tau^2 L^4 = o(\tau/L)$, Theorem 16.2 proves (19.2).

The fractional part belongs to $[0, 1)$ and $\max_x G'(x) = 1/e$, so the displayed expansion gives the upper bound in (19.3). At

$$x_\tau = \tau \lfloor L/\tau \rfloor - L,$$

the fractional part is zero, $x_\tau = O(\tau)$, and $G'(x_\tau) = e^{-1} + O(\tau^2)$. Substitution gives the matching lower bound. Finally, the bijection $z = h_\tau(x)$ identifies the supremum distance for the transformed variable with $\|F_n - J_n\|_\infty$. $\square$

19.1 Uniform central quantiles

The calibration also locates integer quantiles within a bounded number of component-size units. This is finer than the leading Gumbel normalization: the correction to the largest size is generally of order L^2 .

Corollary 19.2 (Quantiles with bounded integer error). *Fix $0 < \eta < 1/2$. For $p \in [\eta, 1 - \eta]$, let*

$$Q_n(p) = \min\{m \in \mathbb{Z} : \mathbb{P}(M_n \leq m) \geq p\}, \quad x_p = -\log(-\log p), \quad \beta_p = -\log p.$$

Uniformly over this interval of probabilities,

$$Q_n(p) = \frac{L + x_p}{\tau} - \frac{(L + x_p)^2 - (L + x_p) - 1 - \beta_p(L + x_p + 1)^2}{4c} + O_\eta(1). \quad (19.5)$$

Proof. Let $y_p = (h_\tau)^{-1}(x_p)$, so $J_n(y_p) = p$. The numbers x_p lie in a fixed compact interval, uniformly in p . On a slightly larger interval,

$$h_\tau(x) - x = O_\eta(\tau L^2), \quad (h_\tau)'(x) = 1 + O_\eta(\tau L^2).$$

In particular, $h_\tau(x_p - 1) < x_p < h_\tau(x_p + 1)$ for all sufficiently large n , uniformly in p . Monotonicity places all y_p in the larger interval, and the derivative bound then gives $y_p - x_p = O_\eta(\tau L^2)$. Applying the mean value theorem to $h_\tau(x) - x$ yields

$$y_p = x_p - \frac{\tau}{4c} \left[(L + x_p)^2 - (L + x_p) - 1 - \beta_p(L + x_p + 1)^2 \right] + O_\eta(\tau^2 L^4). \quad (19.6)$$

Theorem 19.1 gives $\|F_n - J_n\|_\infty \leq C\tau$. On the compact interval just used, J'_n is bounded below by a positive constant depending only on η for all sufficiently large n . Put $u_p = \tau Q_n(p) - L$. The defining inequalities for the integer quantile are

$$F_n(u_p) \geq p, \quad F_n(u_p - \tau) < p.$$

Consequently,

$$J_n(u_p) \geq p - C\tau, \quad J_n(u_p - \tau) < p + C\tau.$$

For large n , both $p - C\tau$ and $p + C\tau$ lie in $[\eta/2, 1 - \eta/2]$. Their J_n -quantiles remain in a fixed compact interval on which the same derivative lower bound applies. Inverting the two inequalities therefore yields

$$J_n^{-1}(p - C\tau) \leq u_p < \tau + J_n^{-1}(p + C\tau),$$

and consequently $u_p = y_p + O_\eta(\tau)$, uniformly in p . Divide (19.6) by τ and use $\tau L^4 \rightarrow 0$. This proves (19.5), including its bounded integer error. $\square$

20 Beyond the lattice scale: the component correction

The universal lattice formula has error $O(\tau/L)$. This scale records the first departure of p_k from the limiting constant c . The stronger hypothesis (14.10) allows that departure to be calculated explicitly.

Theorem 20.1 (Exponential-integral refinement). *Assume (14.10). Define*

$$\tilde{U}_n(x) = U_n(x) - a_* G(x) E_1(\max\{1, L + x\}).$$

Then

$$\|F_n - \tilde{U}_n\|_\infty = O(\tau^{3/2} L^3). \quad (20.1)$$

Moreover, uniformly on $\mathbb{R}$,

$$F_n(x) - U_n(x) = -\frac{a_*}{c} \frac{\tau}{L} G'(x) + O\left(\frac{\tau}{L^2} + \tau^{3/2} L^3\right). \quad (20.2)$$

Thus, if $a_* \neq 0$,

$$\|F_n - U_n\|_\infty \sim \frac{|a_*|}{ce} \frac{\tau}{L}. \quad (20.3)$$

Proof. Work first in (16.1). Monotone sum–integral comparison gives

$$\sum_{k>m} \frac{e^{-\tau k}}{k} = E_1(q) + O\left(\frac{\tau^2 \beta}{L}\right). \quad (20.4)$$

Indeed, the discrepancy at the lower endpoint τm is $O(e^{-\tau m}/m)$, and moving that endpoint to q costs $O(\tau e^{-q}/q)$. Both have the displayed size because $q \asymp L$ and $e^{-q} = \tau \beta/c$.

The residual weight term is bounded by

$$\begin{aligned} \sum_{k>m} \frac{\log(k+1)}{k^2} e^{-\tau k} &\leq C\tau \int_{\tau m}^\infty \frac{e^{-u} \log(1+u/\tau)}{u^2} du + O\left(\frac{e^{-\tau m} \log(m+1)}{m^2}\right) \\ &= O(\tau^2 \beta/L). \end{aligned}$$

Here $\log(1+q/\tau) = O(L)$ on the central range. Together with (16.4), this proves

$$S_0 = \beta + \tau \beta (\theta_n(x) - 1/2) + a_* E_1(q) + O(\tau^2 \beta). \quad (20.5)$$

Exponentiate this identity in the tail-moment formula. The linear contributions are the lattice term in U_n and $-a_* G(x) E_1(q)$. Quadratic products in the exponential are $O(\tau^2)$ after multiplication by $e^{-\beta/2}$. The replacements of S_1, S_2, B_2, B_3 by (16.6), and their products with (20.5), cost $O(\tau^2 L^2)$. All these errors are within $O(\tau^{3/2} L^3)$, the error of Theorem 15.4. This proves (20.1) centrally.

On the right tail $\beta < \tau^2$, both the true upper tail and $1 - U_n$ are $O(\tau^2)$, and $G E_1(q) = O(\tau \beta/q)$. On the left tail $\beta > 16L$, the probability and all explicit polynomial corrections are negligible, as in Theorem 16.2. The cutoff at 1 bounds the additional E_1 factor in the remote region where q is not positive. This proves (20.1) globally. The cutoff is a convention to define the approximation there; it has no effect at the claimed error scale.

Integration by parts gives

$$E_1(q) = \frac{e^{-q}}{q} + O(e^{-q}/q^2).$$

On the central range $q = L - \log \beta \geq L/2$, so

$$\left| \frac{1}{q} - \frac{1}{L} \right| \leq \frac{2|\log \beta|}{L^2}.$$

Boundedness of $G\beta|\log \beta|$ therefore yields

$$G(x) E_1(q) = \frac{\tau}{cL} G'(x) + O(\tau/L^2).$$

The same statement holds on the remote tails by exponential decay and the cutoff convention. Combining it with (20.1) proves (20.2). Since $\|G'\|_\infty = 1/e$ and $\tau^{3/2} L^3 = o(\tau/L)$, the sharp equivalent follows. $\square$

20.1 Why the exponential integral is useful

Repeated integration by parts gives, for each fixed $r \geq 1$,

$$E_1(q) = \frac{e^{-q}}{q} \left(\sum_{j=0}^{r-1} \frac{(-1)^j j!}{q^j} + O_r(q^{-r}) \right), \quad q \rightarrow \infty.$$

Consequently the single E_1 term retains an entire hierarchy of inverse-logarithmic corrections of sizes $\tau/L, \tau/L^2, \dots$. For every fixed r , $\tau^{3/2}L^3 = o(\tau/L^r)$. This assertion is a comparison of fixed error scales; it does not assert convergence of the factorial asymptotic series or uniformity in a truncation order growing with n .

21 Nested-cycle partitions and explicit normalizations

21.1 The combinatorial meaning

A permutation whose cycle lengths are distinct has component EGF

$$P(z) = \prod_{k \geq 1} (1 + z^k/k).$$

For each cycle length k , either no cycle is selected or one labelled cycle is selected; a cycle has EGF z^k/k . A nonempty such permutation is an outer component. Taking a labelled set of those outer components gives $\exp(P(z) - 1)$, exactly (13.1). Thus

$$p_k = [z^k]P(z), \quad k!p_k \in \mathbb{Z}_{\geq 0},$$

and (14.2) is the spectrum of a uniform nested-cycle partition of an n -element label set.

The initial total counts are

$$1, 1, 2, 9, 44, 270, 2064, 17682, 171296, 1867968, 22470840, \dots,$$

in agreement with A308338 [2]. By the credited estimate (13.2),

$$c = a_* = e^{-\gamma} = 0.561459483566885\dots, \quad \sqrt{c} = e^{-\gamma/2} = 0.749306001288449\dots \quad (21.1)$$

All preceding theorems therefore apply.

Corollary 21.1 (Sharp scales for nested-cycle maxima). *For a uniform nested-cycle partition of size n , let M_n be its largest outer component. Then*

$$d_K(F_n, G) \sim \frac{e^{\gamma/2} \kappa \log^2 n}{16 \sqrt{n}}, \quad (21.2)$$

$$d_K(F_n, J_n) \sim \frac{e^{-\gamma/2}}{2e} n^{-1/2}, \quad (21.3)$$

$$\|F_n - U_n\|_\infty \sim \frac{2e^{-\gamma/2}}{e} \frac{1}{\sqrt{n} \log n}, \quad (21.4)$$

$$\|F_n - \tilde{U}_n\|_\infty = O(n^{-3/4} \log^3 n). \quad (21.5)$$

The constant in (21.3) is asymptotically optimal among continuous CDF approximations.

[write] **The corollary from the theorems, 9 October 2026.** With $c = a_*$ from (21.1), $\tau \sim \sqrt{c/n}$ and $L \sim \frac{1}{2} \log n$, the four lines follow from Proposition 17.1 ($\kappa\delta \sim \kappa \log^2 n / (16\sqrt{cn})$), Theorem 19.1 ($\tau/(2e)$), Theorem 20.1 ($|a_*|\tau/(ceL) = \tau/(eL)$) and (20.1); the write rederived each. The constants are $e^{\gamma/2}\kappa/16 = 0.0257742485593614\dots$, $e^{-\gamma/2}/(2e) = 0.1378271365102013\dots$ and $2e^{-\gamma/2}/e = 0.5513085460408053\dots$ (truncations; the manuscript prints none of them). The refined hypothesis (14.10) with $a_* = c$ is exactly Part I's credited estimate (1.4), so this corollary rests on the same external input as Part I and on nothing else.

The largest component has scale $\sqrt{n} \log n$, with fluctuations of order $\sqrt{n}$. The first continuous correction generally changes a fixed central quantile by order $\log^2 n$ in the original integer size. Although this is smaller than the fluctuation scale, it is much larger than one integer unit.

21.2 A two-term explicit saddle

The exact saddle is simple to compute, but an elementary normalization is useful for closed formulas.

Proposition 21.2 (Explicit saddle to the needed precision). *Under (14.10), if $\tau_0 = \sqrt{c/n}$, then*

$$\tau = \tau_0 + \frac{a_*}{2c}\tau_0^2 + O(\tau_0^3 \log^2(1/\tau_0)). \quad (21.6)$$

For nested cycles this is

$$\hat{\tau}_n = \sqrt{\frac{c}{n}} + \frac{c}{2n}, \quad \tau - \hat{\tau}_n = O(n^{-3/2} \log^2 n). \quad (21.7)$$

Using $\hat{\tau}_n$ in place of τ , and $\hat{L}_n = \log(c/\hat{\tau}_n)$ in place of L , preserves the sharp continuous conclusion (21.3), with remainder $O(\hat{\tau}_n/\hat{L}_n)$.

Proof. Summing the refined weight hypothesis gives

$$B_1(t) = \frac{c}{t^2} + \frac{a_*}{t} + O(\log^2(1/t)).$$

Indeed, the residual sum is bounded by a constant times $\sum_{k \geq 1} \log(k+1)e^{-tk}/k = O(\log^2(1/t))$. Substitute $t = \tau_0 + (a_*/(2c))\tau_0^2$; the terms of order τ_0^{-1} cancel. The derivative $-B_2(t) \asymp -\tau_0^{-3}$ near the root, so the mean value theorem gives (21.6).

The relative error in τ is $O(\tau_0^2 L_0^2)$. The induced affine change in the normalized coordinate, on the relevant central range, is $O(\tau_0^2 L_0^3)$. The explicit smooth function $G \circ h$ has bounded weighted derivatives there, and its dependence on τ, L costs the same or less. The two tails are handled by monotonicity and exponential decay, as before. Since $\tau_0^2 L_0^3 = o(\tau_0/L_0)$, the asserted precision is preserved. $\square$

Remark 21.3 (The phase must remain exact). The preceding replacement is justified for continuous approximations. A fractional part is discontinuous. Even a tiny error in the saddle can cross an integer cutoff, changing a lattice approximation by order τ . Therefore (14.7), (20.1), and the exact tail-moment approximation are stated with the exact saddle and the exact cutoff. One can alternatively implement the tail-moment approximation directly at a specified integer m , which avoids evaluating a fractional part numerically.

21.3 A second concrete model: forests of paths

For labelled forests whose connected components are paths, there is one singleton structure and $k!/2$ labelled paths on $k \geq 2$ vertices. Thus

$$p_1 = 1, \quad p_k = \frac{1}{2} \quad (k \geq 2), \quad C(z) := \sum_{k \geq 1} p_k z^k = \frac{z}{2(1-z)} + \frac{z}{2}.$$

This is (14.1) with $c = 1/2$, and (14.10) with $a_* = 0$. A first-order maximum law is already known [10, 11]. The universal lattice approximation here has the improved remainder $O(\tau^{3/2}L^3)$, because the $1/k$ coefficient vanishes. The rational component series also permits numerical checks at very large n without enumerating the objects.

22 Reproducible calculations and their scope

The proofs establish the asymptotic statements. The computations serve two additional purposes: checking the combinatorial model against exact integers, and checking the signs and sizes of the probability corrections against independent coefficient calculations.

22.1 Exact integer and rational checks

Write $B_k^{\text{comp}} = k!p_k$, an integer for nested cycles. The exponential formula gives the recurrence

$$A_0 = 1, \quad A_n = \sum_{k=1}^n \binom{n-1}{k-1} B_k^{\text{comp}} A_{n-k}. \quad (22.1)$$

Restricting the sum to $k \leq m$ gives $A_n^{(\leq m)}$, and

$$\mathbb{P}(M_n \leq m) = A_n^{(\leq m)} / A_n$$

exactly. The supplied code computes p_k from the finite product through the needed degree using rational arithmetic, then uses (22.1) with integers.

The first 22 total counts were checked against the displayed OEIS data. Six exact rational CDF values at $n = 50$ were compared with a full-circle Fourier calculation; the maximum numerical difference was 1.83×10^{-14} . The fractions themselves are retained in `data/verification.json`.

22.2 Numerical Fourier calculations

For larger n , the code evaluates the coefficient ratio (14.13) on a saddle circle. Nested-cycle coefficients are computed by multiplying $1 + z^k/k$ through a finite degree, and the exponentially small weighted tail is cut off. A second implementation uses the rational path-forest generating function

$$C_m(z) = C(z) - \frac{z^{m+1}}{2(1-z)} \quad (m \geq 1).$$

It calculates the same coefficient ratio directly without nested-cycle coefficients.

Comparisons with continuous CDFs include *both ends* of each examined lattice step. Looking only at the value just after a jump can understate the continuous Kolmogorov distance. For nested

cycles every integer threshold in the displayed central window is examined. The larger path-forest cases use a dense normalized grid and additional points near the limiting extrema; their recorded errors are sampled maxima.

Table 1: Nested-cycle numerical errors. Continuous comparisons include both ends of each examined step; lattice comparisons use the exact lattice points. The window is finite and the arithmetic is not interval-certified.

n	Gumbel G	Continuous J_n	Lattice U_n	Refined $\tilde{U}_n$
10^3	8.27×10^{-2}	2.44×10^{-2}	1.59×10^{-2}	1.37×10^{-2}
10^4	3.59×10^{-2}	6.49×10^{-3}	3.80×10^{-3}	3.31×10^{-3}
10^5	1.53×10^{-2}	1.56×10^{-3}	8.28×10^{-4}	7.12×10^{-4}

Table 2: Path-forest numerical errors and ratios to the proved leading equivalents. For the largest sizes, the values are sampled maxima on a dense normalized grid.

n	Raw error	Raw / $(\kappa\delta)$	Calibrated error	Calibrated / $(\tau/(2e))$
10^4	3.69×10^{-2}	1.8618	6.52×10^{-3}	5.0105
10^6	6.66×10^{-3}	1.4170	3.44×10^{-4}	2.6449
10^8	1.09×10^{-3}	1.2716	1.62×10^{-5}	1.2477
10^{10}	1.64×10^{-4}	1.2037	1.32×10^{-6}	1.0140
10^{12}	2.31×10^{-5}	1.1639	1.30×10^{-7}	1.0029

The asymptotic regime is approached slowly in the uncorrected formula. The explicit error profile explains this rather than hiding it: the next term is smaller than the leading error by only a factor of order $1/\log n$. The continuous calibration is substantially more accurate in the experiments, but at moderate n its error need not yet be close to its limiting constant $\tau/(2e)$.

The path-forest implementation was also checked against a separate 60-digit sparse coefficient recurrence at $n = 10^4$. At three cutoffs the maximum difference was 3.34×10^{-16} . Changing 512 to 1024 Gauss–Legendre nodes changed the checked values by at most 2.56×10^{-15} . These numerical checks use floating point and high precision; they do not use directed-rounding interval arithmetic. For nested cycles at $n = 1000$, comparing the finite central Fourier integration with a separate probability recurrence gave maximum discrepancy 1.85×10^{-9} at six checked cutoffs, well below the precision reported in the tables.

22.3 How to reproduce the package

From the package directory:

```
python3 -m pip install -r requirements.txt
python3 code/nested_numerics.py
python3 code/forest_numerics.py --recurrence-check
python3 code/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The package includes all TeX inputs, the compiled PDF, data, verification notes, and a manifest with file hashes. A standard TeX installation with the packages in the preamble is sufficient. No network access is needed to compile the paper once the Python and TeX dependencies are installed.

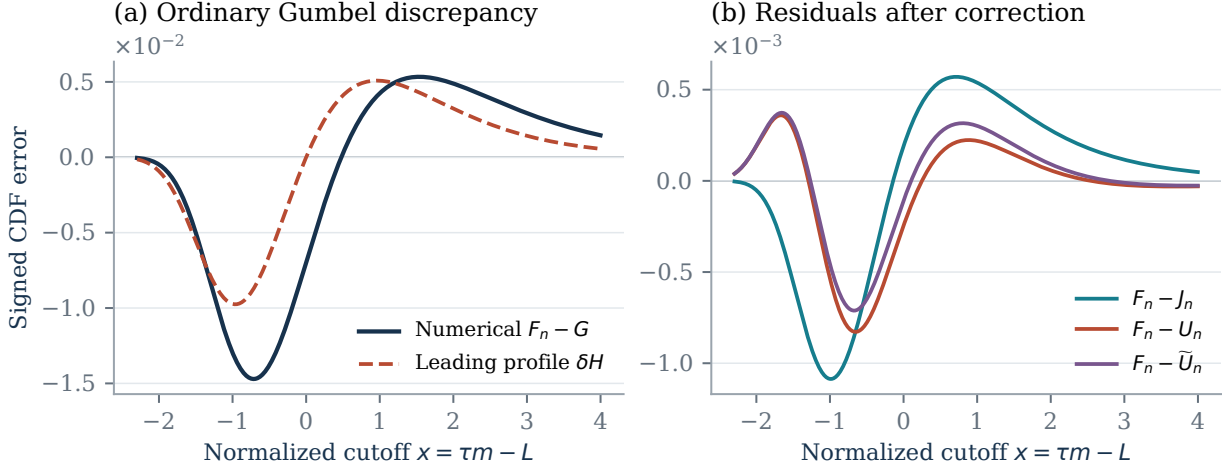


Figure 1: Numerical error profiles for nested-cycle assemblies at $n = 10^5$. The data use actual saddle-circle coefficient ratios. The curves show the signed discrepancy at lattice points; endpoint checks used for continuous error tables are separate. Numerical quadrature supports the asymptotic results but is not the logical basis of their proofs.

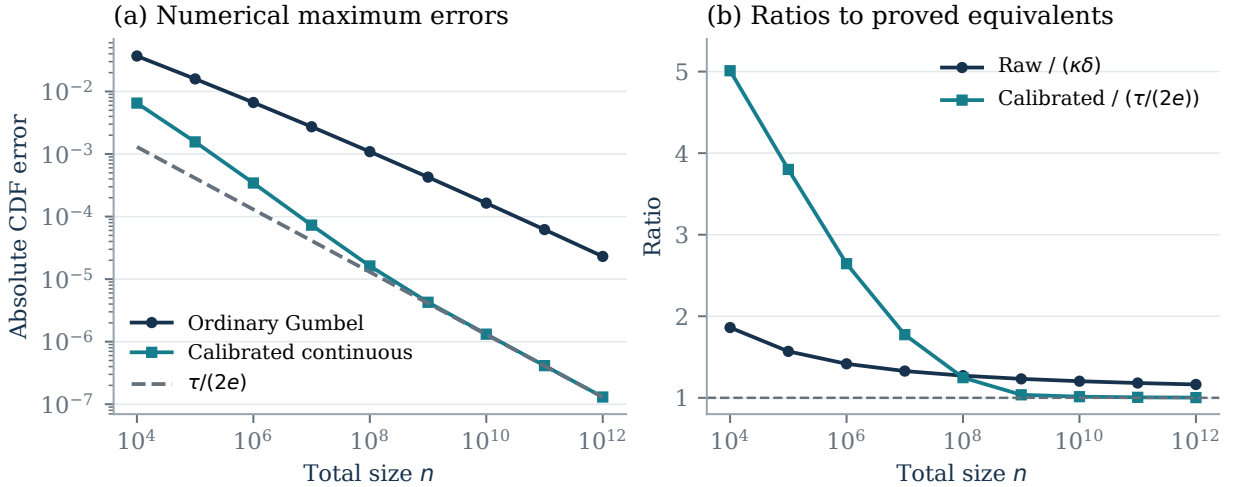


Figure 2: Path-forest calculations over increasing size. The normalized corrected error approaches the sharp constant more quickly than the raw error approaches its own leading equivalent. Larger-size entries are sampled maxima, not interval-certified global suprema.

[write] **The package here, two printed differences, and the write’s reproduction, 9 October 2026.** *Files.* “The supplied code” is `code/02-gumbel-nested_numerics.py`, `code/02-gumbel-forest_numerics.py` and `code/02-gumbel-make_figures.py`; `data/verification.json` and the other data files named in this section are `data/02-gumbel-*`; the compiled PDF and the $\text{\TeX}$ inputs other than the table file are not shipped. The commands above run only on a copy with the delivered names (the README gives the recipe).

Two printed differences. The text’s maximum differences 1.83×10^{-14} (exact against Fourier at $n = 50$) and 1.85×10^{-9} (recurrence against Fourier at $n = 1000$) are recorded in `data/02-gumbel-verification.json` and `02-gumbel-VERIFICATION.txt` as $1.824029208924438 \times 10^{-14}$ and $1.844130930914959 \times 10^{-9}$. The printed values are neither the roundings (1.82, 1.84) nor the truncations of the recorded ones; they are true as upper bounds, and nothing depends on them.

The asymptotic constants are not yet visible. At $n = 10^3, 10^4, 10^5$ the ratios of the errors of Table 1 to their proved leading equivalents are 2.52, 1.87, 1.57 for the raw error against $\kappa\delta$, 5.54, 4.69, 3.58 for the calibrated error against $\tau/(2e)$, and 5.69, 5.92, 5.19 for the lattice error against $\tau/(eL)$. The raw ratio decreases toward 1 as the relative $O(1/L)$ remainder allows; the other two are still far from 1, as the text says (“need not yet be close”). These finite values neither contradict the theorems nor confirm their constants. In the path-forest model ($a_* = 0$) the calibrated ratio reaches 1.0029 at $n = 10^{12}$ (Table 2).

The write’s reproduction (floating point, not certified). The whole distribution of M_n was computed for $n = 10^3, 10^4, 10^5$ by a third method, the incremental product $f_m = f_{m-1} \exp(a_m z^m) \bmod z^{n+1}$ with $\mathbb{P}(M_n \leq m) = f_m[n]/f_n[n]$, in which every term is positive. It gives all four error columns of `data/02-gumbel-nested_errors.csv`, and hence the twelve entries of Table 1, to within 3.0×10^{-10} , 6.6×10^{-15} and 3.1×10^{-14} at the three sizes; at $n = 1000$ it agrees with the intake’s per-cutoff recurrence to every printed digit, so the 3×10^{-10} there is the delivery’s quadrature, inside its own 1.84×10^{-9} . On the same data: the largest atom is 0.0107510, 0.00304167, 0.000911150 (at $m = 135, 576, 2305$, against $L/\tau = 131.9, 573.7, 2304.5$), against $\tau/e = 0.00880258, 0.00276594, 0.000872683$, so $(\max_m \mathbb{P}(M_n = m) - \tau/e)/(\tau^2 L^2)$ is 0.342, 0.262, 0.229, bounded as Theorem 18.1 states; for $p = 0.1, 0.25, 0.5, 0.75, 0.9$ the integer quantile $Q_n(p)$ minus the right side of (19.5) without its $O(1)$ lies in $[-0.69, 1.59]$, $[-0.15, 1.09]$, $[-0.30, 0.61]$, while the correction term itself reaches 26 at $n = 10^5$; and the exact saddle minus $\hat{\tau}_n$ of (21.7) is -4.80×10^{-5} , -2.52×10^{-6} , -1.21×10^{-7} , -5.47×10^{-9} at $n = 10^3, \dots, 10^6$, that is -0.032 to -0.029 times $n^{-3/2} \log^2 n$. The shipped programs, rerun on a copy at the write, reproduce the three CSV files to relative 2×10^{-8} and the six exact fractions exactly (Section 12.3).

[write] **A precision from the independent check, 9 October 2026.** In the quantile sentence of the note above, the largest correction term $|(L + x_p)^2 - (L + x_p) - 1 - \beta_p(L + x_p + 1)^2|/(4c)$ at $n = 10^5$ is 25.48 (at $p = 0.1$); “reaches 26” should read “reaches 25.5”. The three intervals, the atoms, the ratios and the saddle values of the note were confirmed (note at the end of Section 23).

23 Further research questions

The present estimates settle a quantitative problem for a specific, broad weight class. They also isolate several well-defined next steps.

1. **A second correction to the optimal continuous constant.** Determine the next term in $d_K(F_n, G \circ h_\tau)$. The sawtooth and the location of the largest atom interact at a smaller scale, so the correction may depend on a phase subsequence. For nested cycles the $1/k$ component term already gives a systematic contribution of order τ/L .
2. **An optimal nonlinear calibration beyond first order.** The explicit h_τ attains the continuous minimax constant to first order. Can another explicit smooth transformation track half the largest jump through a full expansion? The abstract midpoint interpolant attains the exact minimax error, but uses the unknown finite distribution.

3. **Determine the affine minimax constant E_* .** Theorem 17.2 reduces this question to a two-dimensional weighted uniform approximation problem. A certified equioscillation calculation could identify the optimal center and scale corrections and provide a rigorous numerical value of E_* .
4. **Improve the discrete remainder.** The $O(\tau^{3/2}L^3)$ bound is sufficient to separate the lattice and component scales. It is not claimed optimal. Retaining further terms in the paired integral may yield $O(\tau^2L^r)$ with an explicit polynomial, or a systematic fixed-order expansion. Remainders must still be controlled uniformly on both extreme tails.
5. **Several largest components.** Develop a joint point-process expansion for the largest r components, with fixed r , and identify the conditioning-induced dependence among their limiting Poisson points. The one-cutoff tail moments suggest a finite-dimensional covariance correction.
6. **Broader expansive weights.** For $p_k \sim ck^{\alpha-1}$, $\alpha > 0$, the first-order Gumbel law is known [9]. Determine the analogue of the signed error profile, the best affine rate, and the continuous minimax calibration for $\alpha \neq 1$. Slowly varying factors may compete with conditioning at different scales.
7. **Arithmetic support restrictions.** The present assumption forces eventual positivity at every integer size and hence gives a particularly direct Fourier bound. If weights are supported on one or several residue classes, identify the correct lattice span, admissible sizes, and phase terms. The atom obstruction should change with the span.
8. **Effective finite constants.** Replace the asymptotic O -constants by explicit expressions in c , a bound for $k|p_k - c|$, and an onset of positivity. Combined with interval evaluation of the saddle and Fourier integrals, this would give finite- n certified probability bounds.
9. **Root-of-unity component oscillations.** The fuller expansion of p_k for distinct-cycle-length permutations contains periodic terms [5]. At what order do those oscillations enter the largest-component law, and how do they interact with the outer cutoff phase?
10. **Moment and quantile refinements.** The proved central quantile formula has bounded error in integer size. Extend this to moving probabilities $u = u_n$, and derive comparably precise mean and variance expansions for M_n , with integrable error bounds rather than just a uniform CDF estimate.
11. **Formalization.** A useful first formal target is the analytic transfer from the tail-moment estimate to the explicit profiles and minimax constants. The longer probabilistic target is the common Fourier concentration bound and the paired-integral remainder. Exact count computations should be connected to their combinatorial semantics by a proved checker.

[write] **Further questions of Part II, 9 October 2026.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the manuscript that is not proved in full is stated as an open question; these eleven items are the manuscript’s own, as delivered, and the write labelled them. No claim of the manuscript was found wrong; one printed constant is rounded in its last digit (note after Proposition 17.1) and two printed numerical differences exceed the recorded ones (note at the end of Section 22). *Question 3:* the write’s uncertified minimax computation (weighted error on a grid of $x \in [-4, 25]$, Nelder–Mead) gives $E_* \approx 0.1024413800$ at slope $\alpha \approx 1.0000000$ and shift $b \approx -0.2784645$, with equal weighted errors of alternating signs at $x \approx -1.265$, 0 and 2.164 ; fixing $\alpha = 1$ gives the same value to 7×10^{-10} . It lies between the bounds

0.0402... and $\kappa = 0.3090...$ of (17.5), so the best affine normalization would lower the leading constant by a factor of about 3.02 but not its order. A certified value, and whether $\alpha = 1$ exactly, stay open. *Question 9* is the analogue for M_n of Part I's *Question 3*, and *Question 8* that of Part I's *Question 1*; none of the three is answered.

[write] Independent check of the Part II write: E_* in closed form, 9 October 2026. An independent adversarial check of the Part II write (`aeecf5e9e`) answers *Question 3* of this section. Let $\omega = W(1/e) = 0.2784645427610737951...$ be the real solution of $\omega e^\omega = e^{-1}$. Then

$$E_* = \frac{W(1/e)}{e} = 0.1024413803770050319...,$$

and the infimum in (17.3) is attained only at $\alpha = 1$, $b = -\omega$. *Proof.* Put $\psi_{\alpha,b}(x) = G'(x)(1 - e^{-x} - \alpha x - b)$ and $\psi = \psi_{1,-\omega}$. Since $G'''(x) = G'(x)(e^{-x} - 1)$,

$$\psi'(x) = G'(x)(e^{-x} - 1)(2 + \omega - x - e^{-x}).$$

The function $u(x) = x + e^{-x}$ decreases on $(-\infty, 0]$, increases on $[0, \infty)$ and has $u(0) = 1 < 2 + \omega$, so ψ' vanishes exactly at $x = 0$ and at two points $x_1 < 0 < x_3$ with $u(x_i) = 2 + \omega$. There $1 - e^{-x_i} - x_i + \omega = -1$, so $\psi(x_i) = -G'(x_i) = -\exp(-u(x_i)) = -\exp(-2 - \omega) = -\omega/e$ (because $\exp(-\omega) = e\omega$), while $\psi(0) = \omega G'(0) = \omega/e$. As $\psi(x) \rightarrow 0$ for $x \rightarrow \pm\infty$, $\sup_x |\psi(x)| = \omega/e$, attained with the signs $-, +, -$ at $x_1 < 0 < x_3$. For any (α, b) , $\psi_{\alpha,b} = \psi - G'\ell$ with $\ell(x) = (\alpha - 1)x + b + \omega$. If $\sup |\psi_{\alpha,b}| \leq \omega/e$, then $\ell(x_1) \leq 0$, $\ell(0) \geq 0$ and $\ell(x_3) \leq 0$; as 0 lies between x_1 and x_3 , $\ell(0)$ is a convex combination of $\ell(x_1)$ and $\ell(x_3)$, so all three vanish and $\ell \equiv 0$. Hence $E_* = \omega/e$, attained only at $(1, -\omega)$; in the parametrization $a_n = 1 + \delta\alpha$, $b_n = \delta b$ of the proof of Theorem 17.2 this is $a_n = 1 + \delta$, $b_n = -\omega\delta$. *Numerically:* $x_1 = -1.26514618766573...$, $x_3 = 2.16354785338817...$; a 40-digit solution of the six alternation equations gives $\alpha = 1$ and E_* to all 40 digits, and a linear-programming minimax on two grids agrees. So the write's uncertified value $E_* \approx 0.1024413800$ is low by 3.8×10^{-10} (a grid estimate of a supremum lies below it), its slope $\alpha \approx 1$, shift $b \approx -0.2784645$ and alternation points were right, $\kappa/E_* = 3.0164059...$, and “whether $\alpha = 1$ exactly” is answered: it is. A finite- n certified error bound is still *Question 8*.

The rest of the check. Re-derived with the check's own code: the whole law of M_n at $n = 1000$ in exact integer arithmetic (blocks of equal components added cutoff by cutoff), and at $n = 10^4, 10^5$ by a fourth, signed route, $\mathbb{P}(M_n \leq m) = [w^n]E(w) \exp(-\sum_{k>m} a_k w^k)/[w^n]E(w)$ built downward from $m = n$ (equal to the exact law at $n = 1000$ to 4×10^{-15} ; used for $x \geq -2.5$, below which every error is at most 1.6×10^{-5}). The write's four error values at $n = 1000$ are the exact ones to every printed digit, the delivered ones differ by at most 3×10^{-10} , and at $n = 10^4, 10^5$ all agree to 10^{-13} ; the atoms, the ratios 2.52, 1.87, 1.57; 5.54, 4.69, 3.58; 5.69, 5.92, 5.19, the quantile intervals and the saddle values were reproduced. The six $n = 50$ fractions, the 22 counts and $a(50)$ equal A308338 (revision #21 of 5 October 2026, b-file read again). With mpmath and SymPy: κ (the printed digit is rounded, as the write says), the other stationary value, the ratio $(2 + \sqrt{5})^2 e^{-\sqrt{5}}$, the corollary constants (truncations), the lower bound 0.0402300054..., R_1, R_2, R_3 , the exact geometric tail and its expansion, the derivative (19.4) and its region minima 0.565687 and 0.8125, and that $G'(h_\tau - x)$ is the smooth part of U_n . Also confirmed: the archive and the 18 byte-identical staged files, the pins, the 94 + 72 labels with 19 added (185), the 102 references, and the README's 38 files. One sentence of the note at the end of Section 22 is made precise there. Apart from that sentence and the value of E_* above, no claim of Part II or of the write was found wrong.

A Error scales and notation

Symbol	Meaning
p_k	Assembly component weight
c	Positive limit of p_k
a_*	Coefficient of $1/k$ under the refined hypothesis
τ	Exact solution of $B_1(\tau) = n$
L	$\log(c/\tau)$
a_k	Poisson mean $p_k e^{-\tau k}$
B_j, V	Full size cumulants; $V = B_2$
$S_j(m)$	Cumulant tail above integer cutoff m
Y_n, F_n	Normalized maximum and its CDF
$\beta, q, \theta_n(x)$	$e^{-x}, L + x, \{(L + x)/\tau\}$
G, J_n, U_n	Gumbel, calibrated, and lattice approximations
$\tilde{U}_n$	Lattice approximation with the E_1 term

The scale comparisons used repeatedly are, for any fixed nonnegative integer r ,

$$\tau L^r \longrightarrow 0, \quad \tau^{1/2} L^r \longrightarrow 0.$$

In particular,

$$\tau^{3/2} L^3 = o(\tau/L^r), \quad \tau^2 L^4 = o(\tau/L).$$

These are asymptotic statements with fixed r . They provide no uniform assertion when r grows with n .

The common integration window is $|u| \leq L$, with $u = \sqrt{V} \vartheta$. The common extreme-value window is $\tau^2 \leq e^{-x} \leq 16L$. Keeping these two variables separate avoids conflating the Fourier scale with the extreme-value location.

B Provenance and proof boundaries

The ProveIt source used for the sequence background is `a308338-nested-cycle-assemblies/article.tex`, in the research-report directory recorded in `PROVENANCE.txt`. The inspected source identifiers are:

ProveIt source blob	07022425b718ab5fe3bd332a551baec7cd512379
openai/math commit	adc7f1241b42e322a6451854ab7e4b4c146bf78a

Exact source URLs and the repository snapshot information are provided in the package.

The proof dependencies are:

- The general results use the explicitly stated weight hypothesis, elementary Poisson identities, Fourier inversion, Taylor’s theorem, and elementary sum estimates. Their probability estimates are proved here.
- The nested-cycle application uses the published coefficient estimate (13.2). The refined generating function asymptotics of the previous ProveIt manuscript are not needed as a probabilistic black box.

- The first-order Gumbel limits of prior authors establish context and priority. The quantitative bounds here are derived directly and do not assume an unquoted rate from those sources.
- The code checks finite arithmetic and numerically evaluates distributions. It does not prove the infinite asymptotic estimates. No machine-checked formalization is claimed.

The literature search was directed at expansive assemblies, largest-component Gumbel laws, path forests, and quantitative convergence or second-order corrections. The inspected sources did not supply the explicit profile, affine obstruction, or asymptotically minimax continuous calibration proved here. A broader specialist review remains appropriate before making a publication-level claim of priority.

[write] **The provenance files here, 9 October 2026.** `PROVENANCE.txt` is `02-gumbel-PROVENANCE.txt` in this directory; the blob above is Part I's `article.tex` as it stood before this write (Section 12.1). The proof dependencies listed above agree with the write's reading: no statement of Part I other than the quoted estimate (1.4) enters a proof of this Part.

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- [2] I. Gutkovskiy, *A308338*, OEIS entry dated 20 May 2019. <https://oeis.org/A308338>. Entry inspected 3 October 2026. [Part II: cited as “Gutkovskiy, Ilya and OEIS contributors, A308338: Expansion of e.g.f. $\exp(-1 + \prod_{k \geq 1} (1 + x^k/k))$ ”, entry created 20 May 2019, accessed 8 October 2026; no revision named. Read again at the write (9 October 2026): revision #21 of 5 October 2026, as at Part I's write.]
- [3] M. Riedel, *A392471*, triangle counting nested-cycle assemblies by component count. <https://oeis.org/A392471>. Entry inspected 3 October 2026.
- [4] M. Riedel, *Nested Cycle Partitions: a conjecture*, revised 24 January 2026 author-hosted note, 6 pp. <https://pnp.mathematik.uni-stuttgart.de/iadm/Riedel/papers/ncp-comp2.pdf>. Original 13 January version linked from OEIS: <https://oeis.org/A392471/a392471.pdf>. The nested-cycle specification and component-count discussion were inspected directly.
- [5] P. Flajolet, É. Fusy, X. Gourdon, D. Panario, and N. Pouyanne, *A Hybrid of Darboux's Method and Singularity Analysis in Combinatorial Asymptotics*, *Electronic Journal of Combinatorics* **13**(1) (2006), R103, 35 pp. <https://arxiv.org/abs/math/0606370>. Section 3 and Proposition 1 of the 15 June 2006 version were inspected. The Greene–Knuth attribution is indirect through that work; the book itself was not inspected. [Part II, key FFGPP: cited with DOI 10.37236/1129 for Section 3, equation (22) and Proposition 1; its `02-gumbel-PROVENANCE.txt` records the published version (13 November 2006), an author-hosted PDF and the arXiv preprint as inspected.]
- [6] M. M. Erlihson and B. L. Granovsky, *Limit shapes of Gibbs distributions on the set of integer partitions: The expansive case*, *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* **44**(5) (2008), 915–945. <https://doi.org/10.1214/07-AIHP129>. Published text: <https://numdam.org/item/10.1214/07-AIHP129.pdf>. The assembly weights, expansive scaling, Theorems 4.1 and 4.3, and Corollary 4.5 were checked against the 2007 preprint; relevant published equation numbering and covariance text were also checked.

- [7] M. Erlihson and B. Granovsky, *Reversible coagulation-fragmentation processes and random combinatorial structures: asymptotics for the number of groups*, Random Structures & Algorithms **25** (2004), 227–245. Cited here solely through the explicit historical attribution in [6, Corollary 4.5 and reference 12]; this paper was not inspected directly.
- [8] [Part II] Report 167, *Asymptotics of assemblies of nested cycles: Exact counts and fixed order marked saddle expansions*, research report in the ProveIt repository, 2026; dated 3 October 2026, with repository annotations of 6 October 2026. Inspected Git blob 07022425b718ab5fe3bd332a551baec7cd512379 at commit 2bac1ea07b98, that is, Part I of this report before its second write.
- [9] [Part II] K. Panagiotou and L. Ramzews, *Cluster Statistics in Expansive Combinatorial Structures*, arXiv:2208.00925v1 (1 August 2022). <https://arxiv.org/abs/2208.00925v1>.
- [10] [Part II] M. Bousquet-Mélou and K. Weller, *Asymptotic properties of some minor-closed classes of graphs* (conference version), 25th International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC 2013), Discrete Mathematics & Theoretical Computer Science Proceedings AS (2013), 599–610. <https://doi.org/10.46298/dmtcs.2327>. See Proposition 9.
- [11] [Part II] M. Bousquet-Mélou and K. Weller, *Asymptotic Properties of Some Minor-Closed Classes of Graphs*, Combinatorics, Probability and Computing **23**(5) (2014), 749–795. <https://doi.org/10.1017/S0963548314000303>; arXiv:1303.3836.
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